

INDIAN INSTITUTE OF INFORMATION TECHNOLOGY KOTTAYAM



**CURRICULUM AND SYLLABUS FOR THE COURSES OF B.
TECH./B.TECH (HON)/DUAL DEGREE (B.TECH - MS) PROGRAMME IN
COMPUTER SCIENCE AND ENGINEERING WITH SPECIALIZATION IN
ARTIFICIAL INTELLIGENCE AND DATA SCIENCE**

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B.Tech./B.Tech (Hon)/ B.Tech - MS General Course Structure CSE with Specialization in Artificial Intelligence and Data Science

Semester -I						Semester -II					
Course	Course Name	L	T	P	C	Course	Course Name	L	T	P	C
IMA111	Discrete Mathematics	3	1	0	4	IMA121	Calculus and Linear Algebra	3	1	0	4
ICS111	IT Workshop I	3	1	2	5	IEC121	Digital Design and Electric Circuits	3	1	2	5
ICS112	Computer Programming	3	1	2	5	ICS121	Data Structures I	3	1	2	5
IEC111	Electronic Circuits	3	1	2	5	ICS122	Computer Organization	3	1	0	4
IHS111	Communication Skills	3	0	0	3	ICS123	IT Workshop II	2	1	2	4
IHS112	Foreign Language	1	0	0	1	IHS121	Personality Development	1	0	0	1
IPT111	Physical Training I	0	0	2	0						
Total		16	4	8	23	Total		15	5	6	23
Cumulative Credits at the End of First Year: 46											
Semester-III						Semester -IV					
Course	Course Name	L	T	P	C	Course	Course Name	L	T	P	C
IMA211	Probability, Statistics, and Random Processes.	3	1	0	4	IMA221	Differential Equations and Transforms	3	1	0	4
ICS211	Design and Analysis of Algorithms	3	1	0	4	ICS221	Operating Systems	3	1	0	4
ICS212	Theory of Computation	3	1	0	4	ICS222	Object-Oriented Analysis and Design	3	0	2	4
ICS213	Database Management Systems	2	1	2	4	ICS223	Compiler Design	3	0	2	4
ICS214	IT Workshop III	2	1	2	4	ICS224	Computer Networks	3	0	2	4
ICS215	Data Structures II	1	0	2	2	ICS225	Data Structures III	1	0	2	2
ISC213	Introduction to Cognitive Science	2	0	0	2	IHS221	Fundamentals of Economics	1	0	0	1
IPT211	Physical Training II	0	0	2	0	IHS222	Principles of Management	1	0	0	1
Total		16	5	8	24	Total		18	2	8	24
Cumulative Credits at the End of Second Year: 94											
Semester -V						Semester -VI					
Course	Course Name	L	T	P	C	Course	Course Name	L	T	P	C
CSEXXX	Bouquet Core I	3	0	2	4	CSEXXX	Bouquet Core III	3	0	2	4
CSEXXX	Bouquet Core II	3	0	2	4	CSEXXX	Bouquet Core IV	3	0	2	4
CSSXXX	CS Elective I	3	0	2	4	CSSXXX	CS Elective II	3	0	2	4
IEGXXX	Engineering Elective	3	0	2	4	CSSXXX	CS Elective III	3	0	0	3
IMAXXX	Mathematics Elective	3	0	0	3	ISCXXX	Science Elective	3	0	0	3
IHSXXX	Humanities Elective I	1	0	0	1	IOEXXX	Institute Open Elective I	3	0	0	3
IHSXXX	Humanities Elective II	1	0	0	1	IHP324	Honours Project Phase I (Optional) I				
Total		18	0	8	22	Total		18	0	6	21
Cumulative Credits at the End of Third Year: 137											
Semester-VII						Semester - VIII					
Course	Course Name	L	T	P	C	Course	Course Name	L	T	P	C
CSEXXX	Bouquet Core V	3	0	2	4	CSSXXX	CS Elective V	3	0	0	3
CSEXXX	Bouquet Core VI	3	1	0	4	CSSXXX	CS Elective VI /Industrial Training	3	0	2	4
CSSXXX	CS Elective IV	3	1	0	4	IOEXXX	Institute Open Elective III	3	0	0	3
IOEXXX	Institute Open Elective II	3	0	0	3	IBP421	BTP - II	6	0	0	6
IBP411	BTP-I	6	0	0	6	IHP424	Honours Project Phase II (Optional)				
CSSXXX	Research Course (Optional)					CSSXXX	Research Course (Optional)				
Total		18	2	2	21	Total		15	1	0	16
Cumulative Credits at the End of Fourth Year: 174 (BTech); 174+12=186 (BTech(Hon)); 174+20=194(BTech- MS)											
Semester-IX						Semester -X					
Course	Course Name	L	T	P	C	Course	Course Name	L	T	P	C
IRP511	Research Project	12	0	0	12	IRP521	Research Project	12	0	0	12
Total		12	0	0	12	Total		12	0	0	12
Cumulative Credits at the End of Fifth Year: 218 (BTech-MS)											

B.Tech./B.Tech (Hon)/ B.Tech - MS Detailed Course Structure CSE with Specialization in Artificial Intelligence and Data Science

Semester -I						Semester -II					
Course	Course Name	L	T	P	C	Course	Course Name	L	T	P	C
IMA111	Discrete Mathematics	3	1	0	4	IMA121	Calculus and Linear Algebra	3	1	0	4
ICS111	IT Workshop I	3	1	2	5	IEC121	Digital Design and Electric Circuits	3	1	2	5
ICS112	Computer Programming	3	1	2	5	ICS121	Data Structures I	3	1	2	5
IEC111	Electronic Circuits	3	1	2	5	ICS122	Computer Organization	3	1	0	4
IHS111	Communication Skills	3	0	0	3	ICS123	IT Workshop II	2	1	2	4
IHS112	Foreign Language	1	0	0	1	IHS121	Personality Development	1	0	0	1
IPT111	Physical Training I	0	0	2	0						
Total		16	4	8	23	Total		15	5	6	23
Cumulative Credits at the End of First Year: 46											
Semester-III						Semester -IV					
Course	Course Name	L	T	P	C	Course	Course Name	L	T	P	C
IMA211	Probability, Statistics, and Random Processes.	3	1	0	4	IMA221	Differential Equations and Transforms	3	1	0	4
ICS211	Design and Analysis of Algorithms	3	1	0	4	ICS221	Operating Systems	3	1	0	4
ICS212	Theory of Computation	3	1	0	4	ICS222	Object-Oriented Analysis and Design	3	0	2	4
ICS213	Database Management Systems	2	1	2	4	ICS223	Compiler Design	3	0	2	4
ICS214	IT Workshop III	2	1	2	4	ICS224	Computer Networks	3	0	2	4
ICS215	Data Structures II	1	0	2	2	ICS225	Data Structures III	1	0	2	2
ISC213	Introduction to Cognitive Science	2	0	0	2	IHS221	Fundamentals of Economics	1	0	0	1
IPT211	Physical Training II	0	0	2	0	IHS222	Principles of Management	1	0	0	1
Total		16	5	8	24	Total		18	2	8	24
Cumulative Credits at the End of Second Year: 94											
Semester –V						Semester -VI					
Course	Course Name	L	T	P	C	Course	Course Name	L	T	P	C
CSE311	Artificial Intelligence	3	0	2	4	CSE321	Embedded Systems Design	3	0	2	4
CSE312	Software Architecture: Principles and Practices	3	0	2	4	CSE322	Cloud Computing	3	0	2	4
CSSXXX	Parallel and Distributed Computing / Cyberphysical Systems and IoT	3	0	2	4	CSS321	Data Warehousing and Data Mining	3	0	2	4
IEGXXX	Digital Signal Processing/ Quantum Computing for Engineers: Theory and Applications	3	0	2	4	CSS322	Machine Learning	3	0	0	3
IMAXXX	Optimization techniques for Data Science/ Linear Algebra for Data Science	3	0	0	3	ISCXXX	Introduction to Network Science Analytics	3	0	0	3
IHS311	Human Resource Management	1	0	0	1	IOEXXX	Introduction to DevOps and Microservices	3	0	0	3
IHS312	Financial Management and Accounting	1	0	0	1	IHP324	Honours Project I (Optional)				
IHS313	Operations and Supply Chain Management	1	0	0	1						
Total		18	0	8	22	Total		18	0	6	21
Cumulative Credits at the End of Third Year: 137											
Semester-VII						Semester - VIII					
Course	Course Name	L	T	P	C	Course	Course Name	L	T	P	C
CSE411	Computer Vision: Principles and Processes	3	0	2	4	CSSXXX	Applied Predictive Analytics / Computer Systems for Machine Learning/ Scientific Machine Learning/ Time Series Analytics Edge Computing	3	0	0	3
CSE413	Speech and Natural Language Processing	3	0	2	4	CSSXXX	Machine Learning Operations(MLOps)/ Graph Analytics for big data	3	0	2	4
CSS411	Cryptography and Network Security	3	0	2	4	IOE421	Large Language Models	3	0	0	3
IOEXXX	Data handling analysis and visualization/ Streaming data analytics/ Dependable and secure AI/ Pattern recognition/ Deep Learning	3	0	0	3	IBP424	B.Tech Project Phase II	6	0	0	6
IBP414	B.Tech Project Phase I	6	0	0	6	IHP424	Honours Project Phase II (Optional)				
CSSXXX	Research Course (Optional)					CSSXXX	Research Course (Optional)				
Total		18	0	6	21	Total		15	0	2	16
Cumulative Credits at the End of Fourth Year:174 (B. Tech); 174+12=186 (B. Tech (Hon)); 174+20=194 (B. Tech- MS))											
Semester -IX						Semester -X					
Course	Course Name	L	T	P	C	Course	Course Name	L	T	P	C
IRP511	Research Project	12	0	0	12	IRP521	Research Project	12	0	0	12
Total		12	0	0	12	Total		12	0	0	12
Cumulative Credits at the End of Fifth Year: 218 (B. Tech-MS)											

Remark: To meet the minimum requirement of 186 credits for qualifying the B.Tech (Hon)Degree, students may take two additional projects of 6 credits each and, to meet the requirement of 218 credits for B.Tech-MS, students may take two additional projects of 6 credits each, two 4 credit research courses and 24 credit research project in addition to 174 credits requirement of B.Tech Degree.

B.TECH./B.TECH (HON)/DUAL DEGREE (B.TECH - MS) PROGRAMME

Sl. No.	Course Description	Minimum Credits Requirement			Period
		B.Tech	B.Tech(Hon)	B.Tech-MS	
1	Institute Core/Foundation Courses	94	94	94	Semester I to IV
2	Bouquet Core Courses	24	24	24	Semester V to VIII
3	Stream Electives	18	18	18	Semester V to VIII
4	Institute Open Electives	9	9	9	Semester V to VIII
5	Humanities Electives	3	3	3	Semester V to VIII
6	Science Electives	3	3	3	Semester V to VIII
7	Maths Elective	3	3	3	Semester V to VIII
8	Engineering Elective	4	4	4	Semester V to VIII
9	Any other Electives/Industrial Training	4	4	4	Semester V to VIII
10	B.Tech Projects	12	12	12	Semester VII to VIII
11	Honours Project	-	12	12	Semester VI to VIII
12	Research Courses	-	-	8	Semester VI to VIII
13	Research Project	-	-	24	Semester IX to X
Total Credits required for Successful Completion		174	186	218	
Minimum CGPA required for Successful Completion		5.5	8.0	8.0	

SEMESTER I

IMA111 Discrete Mathematics [3-1-0-4]

Course Objectives

- To extend student's Logical and Mathematical maturity and ability to deal with abstraction
- To introduce most of the basic terminologies used
- To explain and apply the basic methods of discrete mathematics
- To able to write clear, concise and correct mathematics proofs.
- To solve counting problems involving permutations and combinations and apply Pigeon hole principle
- To understand the basics of graph theory and group theory

Course Outcomes

- Have knowledge of the concepts needed to test the logic of a program.
- Have an understanding in identifying structures on many levels.
- Be aware of a class of functions which transform a finite set into another finite set which relates to input and output functions in computer science.
- Be able to apply basic counting techniques to solve combinatorial problems
- Acquire ability to describe computer programs in a formal mathematical manner.

Syllabus

Logic: Propositions, negation, disjunction and conjunction, implication and equivalence, truth tables, predicates, quantifiers, rules of inference, methods of proof.

Set theory: definition and simple proofs in set theory, Inductive definition of sets and proof by induction, inclusion and exclusion principle, relations, representation of relations by graphs, properties of relations, equivalence relations and partitions, partial orderings, Functions: mappings, injection and surjections, composition of function, inverse functions, special functions, recursive function theory.

Elementary combinatorics: Counting techniques, pigeon-hole principle, recurrence relation, generating functions. Graph theory: Elements of graph theory, Euler graph, Hamiltonian path, trees, tree traversals, spanning trees.

Algebra: groups, Lagrange's theorem, homomorphism theorem, rings and fields, structure of the ring $\mathbb{Z}_n$ and the unit group $\mathbb{Z}_n^*$, lattice.

Textbooks/References

1. Kenneth H. Rosen, Discrete Mathematics and Its Applications, Seventh Edition, McGrawHill, 2017.

2. Norman L. Biggs, Discrete Mathematics, Oxford University Press, Second Edition, 2003.
3. J.P. Tremblay, R. Manohar, Discrete Mathematical Structures with applications to Computer Science, McGraw Hill, 2017.
4. K.A. Ross, C.R. B. Wright, Discrete Mathematics, 5th Edition, Pearson, 2003.
5. P. B. Bhattacharya, S. K. Jain, S. R. Nagpaul, Basic Abstract Algebra, Second Edition, Cambridge University Press, 2003.
6. J.A. Gallian, Contemporary Abstract Algebra, Ninth Edition, Cengage Learning, 2017.

ICS111 IT Workshop [3-1-2-5]

Course Objectives

- To extend student's knowledge in basics of computers and networks.
- To enable students to understand the internals of systems, which includes hardware parts, software and its installation procedure, basic Linux commands.
- Students would also be getting familiar with various networking terminologies and its components.
- They would be also studying to setup and develop a dynamic web application.

Course Outcomes

- Have a knowledge of the various hardware components.
- Understand Linux commands and shell scripting.
- Be aware of basic networking concepts, devices and its functionality.
- Be aware of the basic web development scripting languages like HTML, CSS, XML, and JavaScript.

Syllabus

Computer Hardware – Prerequisite, Computing Agents CPU, Memory, Motherboard - Computer Peripherals - I/O devices, Storage devices, Interface cards – Buses – Firmware - Boot process - Computer System Software - Operating Systems, Unix/Linux commands, Shell scripting. Computer Communications – LAN, MAN, WAN, Client/Server networks, Peer-to-Peer networks, Topologies, Basics of IP addresses, DNS, Routers, Internet, WWW, Web servers.

Web Design - Basics of HTML, CSS and JavaScript, Dynamic Webpage creation, Angular JS: The basics of AngularJS, Introduction MVC, Data Binding, Filters and modules, Directives, Working with Forms, Validation, Angular JS animation, Deployment considerations.

Lab Practice:

- Computer Hardware – Familiarization CPU Box, Mother board, CPU & Chip-set, Interface cards, Card slots, Hard disk, Cables, SMPS, NIC, Various ports, etc.

- Computer Peripherals - I/O Devices. Storage devices, Interface cards – Buses – Firmware - Boot process - Writing/formatting media.
- Computer System Software - Operating Systems, Windows, Linux, Commands Unix/Linux commands, Shell scripting. Client/Server networks, Peer-to-Peer networks, LAN, WAN, MAN.
- Familiarization of - Basics of TCP/IP, IP addresses, DNS, Routers, Internet, WWW, FTP, Email servers, Web servers. Web Design - Basics of HTML, CSS, XML, Java Scripting

Textbooks/References

1. Kevin Wilson, Exploring Computer Systems: The Illustrated Guide to Understanding Computer Systems, Hardware & Networks, Volume 6 of Exploring Tech, Elluminet Press, 2019.
2. Craig Zacker, John Rourke, PC Hardware: The Complete Reference, McGraw Hill Education, 1st edition, 2017.
3. Shotts, W, The Linux command line: a complete introduction. 2nd Ed., No Starch Press, 2019.
4. Alan Clements, Principles of Computer Hardware, Oxford University Press India, Fourth Edition, 2013.
5. Robbins, J. N. (2012). Learning web design: A beginner's guide to HTML, CSS, JavaScript, and web graphics. "O'Reilly Media, Inc."
6. Meloni, Julie C., Kyrnin, Jennifer. HTML, CSS, and JavaScript All in One, Sams Teach Yourself. United Kingdom: Pearson Education, 2018.
7. Duckett, Jon. Web design with HTML, CSS, JavaScript and jQuery set. Vol. 1. IN: Wiley, 2014.
8. Andrew Grant, Beginning AngularJS. Apress, 2014.
9. Angular: Up and Running: Learning Angular, Step by Step, O'Reilly; 1st edition, 2018.

ICS112 Computer Programming [3-1-2-5]

Course Objectives

- To introduce the problem-solving processes and techniques
- To teach computer programming
- To use C & C++ for solving the problems

Course Outcomes

- Students learn how to write the sequence of operations in solving a problem
- Students learn to translate the problem solving steps to a program
- Students learn the use of programming language for solving real world problems on a computer

Syllabus Basics of computers: software/ systems, Programming- Introduction, Problem solving, Introduction, Problem solving techniques: definition of problems, solutions, top-down approach, breaking problem in to sub-problems.

Algorithms: - writing the steps required solving problems, representing algorithms as flowchart, translating to procedure/ functions. Modularity

Example problems: computation of factorial, sine, Mod arithmetic computation of quotient/remainder, solving factorial through recursion, etc. Object technology- introduction, C + + data types/ scope rules, C + + control statements, Example problems/ program, Example problems/ program, Character handling, Pointers, functions, Classes and objects, Classes and objects.

Textbooks/References

1. D. A. Patterson and J. L. Hennessy, Computer Organisation and Design: The Hardware/ Software Interface, Fourth Edition, Morgan Kaufman, 2009.
2. V. P. Heuring and H. F. Jordan, Computer System Design and Architecture, Prentice Hall, 2003.
3. J. L. Hennessy and D. A. Patterson, Computer Architecture: A Quantitative Approach, Fifth Edition, Morgan Kaufman, 2011.
4. Carl Hamazher, Zvonko Vranesic and Safwat Zaky, Computer Organization, Fifth Edition, McGraw Hill, 2002.

IEC111 Electronic Circuits [3-1-0-4]

Course Objectives

- To impart the basic concepts of semiconductor devices.
- To make students capable of analyzing the operation of various electronic circuits.
- To develop knowledge for designing simple analog electronic circuits using discrete and integrated components.

Course Outcomes

- Ability to analyze PN junctions in semiconductor devices under various conditions.
- Ability to design and analyze simple rectifier/clipper/clamper circuits using diodes.
- Ability to design and analyze amplifier circuits using BJTs and MOSFETs.
- Ability to simulate simple electronic circuits using softwares such as LTSpice/PSpice.

Syllabus

Semiconductors: Intrinsic semiconductors, doped semiconductors, current flow in semiconductors, drift current, diffusion current, the pn junction, physical structure, semiconductor diode, diode equation, diode characteristics, static and dynamic resistances, diode equivalent models: ideal diode model,

piecewiselinear model, constant voltage-drop model, diode applications: rectifiers, clipping and clamping circuits, zener diodes as regulators.

Bipolar Junction Transistors (BJTs): Introduction to transistors, transistor construction, operation, NPN and PNP transistors, transistor voltages and currents, transistor characteristics. Common-base and common-emitter transistor configurations, biasing BJTs: fixed-bias, emitter-stabilized bias, voltage-divider bias, dc bias with voltage feedback, emitter follower bias, current mirror, BJT as amplifiers, voltage gain, current gain, input resistance and output resistance using re model, BJT as a switch.

Field Effect Transistors: Introduction, MOSFET, device structure and physical operation, regions of operation, current-voltage characteristics, body effect, biasing of MOSFET. **Optoelectronic devices:** photodiodes - current and voltage in an illuminated junction, photodetectors. **Light Emitting Diode (LED).**

Textbooks/References

1. Sedra A. and Smith K. C, "Microelectronic Circuits", Oxford University Press, Seventh Edition, 2015..
2. Robert. L. Boylestad, Louis Nashelsky, "Electronic Devices and Circuits Theory", Pearson Education, Eleventh Edition, 2015.
3. David A Bell, "Electronic Devices and Circuits", Oxford University Press, Fifth Edition, 2008.
4. Jacob Millman and Christos. C. Halkias, "Electronic Devices and Circuits", Mc. Graw Hill, Fourth Edition, 2015.
5. Ben. G. Streetman, Sanjay Kumar Banerjee, "Solid State Electronic Devices", 7th Edition, Pearson Education, 2016, ISBN 978-93-325-5508- 2.
6. Donald A Neamen, Dhruves Biswas, "Semiconductor Physics and Devices", 4th Edition, MCGraw Hill Education, 2012, ISBN 978-0-07-107010-2.
7. S. M. Sze, Kwok K. Ng, "Physics of Semiconductor Devices", 3rd Edition, Wiley, 2018.

IHS111 Communication Skills [3-0-0-3]

Course Objectives

- The objective of the course is to improve the English communication skills of First Semester B. Tech students who had just passed out of their Senior Secondary classes. This is challenging since the class included students from various parts of the country speaking various mother tongues.
- The syllabus is designed to give importance to essential grammar, as well as reading, writing and speaking skills. Based on this, work in class consisted of teaching grammar, interspersed with written exercises, reading practice, reading comprehension, business letter writing, report writing, training in preparing CV for job applications, group discussion and extempore speaking. The classes

were rounded off with some training in the so called "soft skills".

Course Outcome

- At the end of the sessions, improvement in English language ability was noted in most of the students. A large number showed very good improvement, while even the least competent registered some improvement. Under the circumstances, the objective of the class would appear to have been achieved.

Syllabus

Communication, verbal and non-verbal, Conversation: formal and informal, prepared and extempore. English: British/American/Indian, Vocabulary development: reading, use of dictionaries. Expression: writing, Pronunciation: phonetics, use of phonetic dictionaries, speaking. English grammar: Basics: Parts of speech: Noun, Pro-noun, Adjective, Verb, Adverb, Preposition, Conjunction, Interjection. Verb: Tenses. Sentence structure: S+V.Concord: Subject-Verb agreement.

Reported speech,Active and passive voice, Tag questions, Confusing words and expressions,Synonyms and antonyms,Idioms and phrases, Common errors in English,Punctuation.Writing skills: Letters: Formal/Informal, Reports,CV. Comprehension: Listening/Reading/Making notes/ Summarising, Interview skills, Group discussion,Soft-skills.

Textbooks/References

1. A.J. Thomson & A.V. Martinet.A Practical English Grammar. Delhi: OUP.
2. George Yule.xford Practice Grammar: Advanced.Oxford:OUP.
3. Raymond Murphy.Essential English Grammar. Delhi: Cambridge University Press.
4. Matthew Monippally.The Craft of Business Letter Writing. New Delhi: Tata McGraw-Hill.

IHS112 Foreign Language [3-0-0-3]

Course Objectives

- To introduce the basics of the German language and its grammar to the students.
- To develop the four fundamental language skills at the initial level: listening, speaking, reading, and writing.
- To cover the essentials of the German language, such as the German alphabet and pronunciation, core grammar rules, and simple, everyday vocabulary.
- To provide a basic understanding of the German language and the culture of German-speaking countries (Germany, Austria, Switzerland, etc.).

Course Outcomes

Upon successful completion of this course, students will be able to:

- Introduce themselves and others, and ask and answer basic personal questions (e.g., about where they live, people they know, and things they have).
- Understand and use familiar, everyday expressions and very simple sentences aimed at satisfying concrete needs (e.g., ordering food, asking for directions).
- Interact in a simple way, provided the other person talks slowly and clearly and is prepared to help.
- Read and understand the main points in short, simple texts like public notices, posters, or short personal messages.
- Write a short, simple text, such as a postcard with holiday greetings, or fill in forms with personal details like their name, nationality, and address.
- Develop a basic awareness of German culture and daily life.

Syllabus

Introduction and Basics: The German Alphabet and Pronunciation (Das deutsche Alphabet und Aussprache): Learning the sounds of German letters and letter combinations. Greetings and Introductions (Begrüßungen und Vorstellungen): Formal and informal greetings, introducing oneself and others. Numbers, Time, and Dates (Zahlen, Uhrzeit und Datum): Numbers from 0 to 100, telling time, and mentioning dates.

Basic Grammar and Sentence Structure: Nouns, Gender, and Articles (Nomen, Genus und Artikel): Understanding that German nouns have a gender (masculine, feminine, neuter). Using definite and indefinite articles (*der/die/das, ein/eine/ein*) in the Nominative case. Simple Sentence Structure: Forming basic statement sentences and questions (W-Fragen like *Wer? Was? Wo?* and Ja/Nein questions). Verb Conjugation (Present Tense) (Konjugation im Präsens): Conjugating regular verbs and important irregular verbs like *sein* (to be), *haben* (to have), and *sprechen* (to speak). A1 Connectors, Basic writing.

Building Vocabulary and Conversations: Topics: The Family (*die Familie*), Hobbies (*die Hobbys*), Food and Drinks (*Essen und Trinken*), and Professions (*Berufe*). Grammar: The Accusative case for direct objects. Negation with *nicht* and *kein*. Possessive Articles (*Possessivartikel*): my, your, his, her, etc. (e.g., *mein Buch, deine Tasche*).

Expanding Skills: Talking about the Past: Introduction to the Present Perfect tense (*Perfekt*) to talk about past events (e.g., "I have learned German"). Separable Verbs (*Trennbare Verben*): Understanding verbs like *anrufen* (to call) or *einkaufen* (to shop). Imperative sentence formation.

Textbooks/References

1. *Netzwerk neu A1* (Kursbuch und Arbeitsbuch) by Klett Sprachen.
2. *Netzwerk neu A1* (Vocabulary) by Klett Sprachen.
3. *Grammatik aktiv A1-B1* by Cornelsen.

SEMESTER II

IMA121 Calculus and Linear Algebra [3-1-0-4]

Course Objectives

- To Study the basic topological properties of the real numbers
- Have the knowledge of the sequence of real numbers and convergence.
- Studying the notion of continuous functions and their properties.
- To gain an understanding of the linear system of equations
- To get introduced to the fundamental concepts of vector spaces
- To impart the basics of linear transformation, orthogonalization, basis, dimensions and eigenvalues.
- To provide the knowledge to apply the concepts of linear algebra in engineering applications

Course Outcomes

- Have a good knowledge of the mathematical concepts in real analysis
- Be able to prove statements and to formulate precise mathematical arguments.
- To solve the problems related to linear systems and matrices
- To apply the knowledge of linear transformation, orthogonal projections, orthonormalization and Least-square solutions in engineering applications.

Syllabus

Calculus: Real Numbers, Properties of Real Numbers, Limit, Continuity, Discontinuity, Uniform continuity, Differentiability, Rolle's theorem, Mean Value theorem, Riemann Integration, fundamental theorem of calculus.

Sequences and Series: Convergence and limit laws, Finite and infinite series, Sums of non-negative numbers, Absolute and conditional convergence of an infinite series, tests of convergence;

Function of several variables: Limit, Continuity, Partial derivatives, Local maxima and local minima, Saddle point, Hessian Matrix.

Vector Calculus: Parameterised curves in space, Arc length, Tangent and normal vectors, Line integral, Gradient, Directional derivatives, Tangent plane and normal vector, Vector field, Divergence, Curl, Parameterised surface, Surface integral, Integral theorems (statements only): Green's Theorem, Stokes' theorem, Gauss' divergence theorem. Co-ordinate system: Polar, spherical, cylindrical.

Linear Algebra: System of linear equations, Row reduced echelon matrices, Rank of a matrix. Definition of a linear

vector space and examples; linear independence of vectors, basis and dimension, Subspaces; Linear transformations, Matrix representation of Linear transformation, Inner product, norms, orthogonal basis, Gram-Schmidt orthogonalization process, Eigen vectors of a matrix and matrix diagonalization, Applications.

Textbooks/References

1. G. Bartle and D. R. Sherbert, Introduction to Real Analysis, Fourth Edition, Wiley, 2011.
2. T. M. Apostol, Calculus, Volume I, Second Edition, Wiley, 2007.
3. Gilbert Strang, Linear Algebra and Its Applications, 5 edition, Wellesley-Cambridge Press/Siam, 2016
4. K. Hoffman and R. Kunze, Linear Algebra, 2 edition, PHI, 2009
5. Erwin Kreyzig, Advanced Engineering Mathematics, Tenth Edition, Wiley, 2015.

IEC121 Digital Design and Electric Circuits [3-1-2-5]

Course Objectives

The primary objective of this course is to provide the student with the fundamental concepts and skills necessary to analyze and design combinational and sequential logic circuits. The course explains the basics of analog and digital logic circuits. It also introduces the student a hardware description language and its application to the design of combinational, sequential and simple digital systems. The material covered in the lecture is reinforced through practical experience in the associated lab together with the use of Verilog HDL to synthesize logic circuits.

Course Outcomes

- Understand the fundamentals of analog and digital circuits.
- Analyze and design a circuit of logic gates that have the desired relation between the input and output terminals.
- Understand the logic properties of flip flops.
- Analyze and design counters, registers, and similar circuits.
- Implement combinational and sequential circuits using a hardware description language.

Syllabus

Introduction - Analog and Digital circuits – Kirchhoff's Laws, Superposition Theorem, Thevenin's and Norton's Theorems; Review of Number Systems - Number systems and conversions- decimal, binary, 1's and 2's complements, hexadecimal, octal etc. Logic gates-NOT, AND, OR, XOR, XNOR, Universal gates, timing diagrams.

Boolean algebra-DeMorgans theorems, SOP and POS forms. Karnaugh Maps-to simplify Boolean expressions, truth table functions. Combinational Logic-Analyze basic combinational

logic circuits, design a combinational logic circuits for a given truth table. Functions of Combinational logic-comparators, adders, code converters, multiplexers, de-multiplexers.

Sequential Circuit Design - Flip-Flops and Latches. SR, D, and JK Flip-Flops. Edge-triggered and Master-Slave Flip-Flops, Excitation table. Counters – Design of asynchronous and synchronous counters. Timing diagrams up/down counters. Shift Registers – data movements in shift registers. SISO, SIPO, PISO, PIPO shift registers.

Memory and programmable logic – RAM, Memory decoding, ROM, PLA, PAL, sequential programmable devices, overview of logic design using Verilog HDL, Basic concepts, Modules, Ports.

Lab Practice

- Familiarization of Logic Gates.
- Design of Combinational Logic Circuits – Comparators, Adders, Code Converters, Multiplexers, Demultiplexers etc.
- Familiarization of Flip-Flops and Latches. SR, D, and JK Flip-Flops. Edge-triggered and Master-Slave Flip-Flops.
- Design of Sequential Logic Circuits, Design of Counters, Asynchronous Counters, Synchronous counters. Shift Registers.
- Simple Verilog HDL programs.

Textbooks/References

1. Floyd, Digital Fundamentals, McGraw Hill, Tenth Edition, 2011.
2. Morris Mano, Digital Circuits and Logic Design", PHI Publication, Fifth Edition, 2015.

ICS121 Data Structures I [3-1-2-5]

Course Objectives

- Define and describe simple data structures like arrays, linked lists, trees and graphs
- Design and specify algorithms for searching and sorting, and those associated with the above data structures
- Analyse simple algorithms, like sorting and searching using mathematical tools, like formulation and solving of recurrences, asymptotic analysis and probabilistic analysis
- Analyse application problems and abstract them to formulate solutions involving data structures and algorithms

Course Outcomes

- Students learn to define operations of data structures like arrays, linked lists, trees and graphs
- Students learn to design and specify algorithms involving above types of data structures

- Students learn to analyze simple algorithms and solve recurrences, asymptotic analysis and probabilistic analysis
- Students learn to analyze application problems and abstract them to formulate solutions involving data structures and algorithms.

Syllabus

Introduction- Algorithm Analysis, Finding Complexity. Fundamental data structures - ListSorted Lists, Double Linked Lists, Skip list.

Stack & Queue application.- Celebrity problem, histogram rectangular area problem.

Binary Trees - Insertion and Deletion of nodes, Tree Traversals, Polish Notations, Red Black Trees, B-Trees, Heaps, Priority Queues.

Optimal binary search tree, Application problems on Optimal binary search Tree.

Sorting - Bubble, Selection, Insertion, Merge Sort, Quick Sort, Radix Sort, Heap sort. Searching.

Hashing- Application problems on hashing.

Graphs- Shortest path algorithms, Minimum Spanning Trees, BFS, DFS.

Textbooks/References

1. Clifford A Shaffer, Data Structures and Algorithm Analysis, Edition 3.2 (Java Version), 2011.
2. Michael T. Goodrich, Roberto Tamassia, Michael H. Goldwasser. Data Structures And Algorithms In Java TM Sixth Edition, Wiley Publishers, 2014.
3. Mark Allen Weiss Data Structures And Algorithm Analysis In Java, Third Edition, 2012.
4. Robert L. Kruse, Data Structures And Program Design In C++, Pearson Education, Second Edition, 2006.
5. Ellis Horowitz, Fundamentals of Data Structures in C++, University Press, 2015.
6. Ajay Agarwal, Data Structure through C, A Complete Reference Guide, Cyber Tech Publications, 2005

ICS122 Computer Organization [3-1-0-4]

Course Outcomes

- To understand the basics of computer hardware and how software interacts with computer hardware.
- To analyze and evaluate the performance of computers.
- Understand basics of Instruction Set Architecture (ISA) – RISC.
- To understand how computers, represent and manipulate data.

- To understand how computer, perform arithmetic operations, how they are optimized and made to run faster.
- To understand how the memory management takes place in a computer system.
- To understand what is pipelining, and the design concepts involved.
- Design a simple computer with hardware design including data format, instruction format, instruction set, addressing modes, bus structure, input/output, memory, Arithmetic/Logic unit, control unit, and data, instruction and address flow.

Course Objectives

- This course will introduce to students the fundamental concepts underlying modern computer organization and architecture.
- Students should be able to know the overall working of a computer.
- Students should be able to get a detailed understanding of the design principles involved in developing a computer.
- They should know the representation of data, how programs are represented, executed and how programs manipulate and operate on data.
- They should also be able to appreciate how the memory organization is done and how to organize memory for faster execution of programs.
- Students should also be able to appreciate the concepts in pipelining.

Syllabus

Computer abstraction and technology: Basic principles, hardware components, Measuring performance: evaluating, comparing and summarizing performance. Instructions: operations and operands of the computer hardware, representing instructions, making decision, supporting procedures, character manipulation, styles of addressing, starting a program.

Computer Arithmetic: signed and unsigned numbers, addition and subtraction, logical operations, constructing an ALU, multiplication and division, floating point representation and arithmetic, Parallelism and computer arithmetic.

The processor: building a data path, simple and multi-cycle implementations, microprogramming, exceptions, Pipelining, pipeline Data path and Control, Hazards in pipelined processors.

Memory hierarchy: caches, cache performance, virtual memory, common framework for memory hierarchies Input/output: I/O performance measures, types and characteristics of I/O devices, buses, interfaces in I/O devices, design of an I/O system, parallelism and I/O. Introduction to multicores and multiprocessors

Textbooks/References

1. D. A. Patterson and J. L. Hennesy, Computer Organisation and Design: The Hardware/ Software Interface, Fourth Edition, Morgan Kaufman, 2009.
2. V. P. Heuring and H. F. Jordan, Computer System Design and Architecture, Prentice Hall, 2003
3. Computer Architecture: A Quantitative Approach, Fifth Edition, Morgan Kaufman, 2011.
4. Carl Hamazher, Zvonko Vranesic and Safwat Zaky, Computer Organization, Fifth Edition, McGraw Hill, 2002.

ICS123 IT Workshop II [2-1-2-4]

Course Objectives

- To introduce the object-oriented problem-solving processes/ techniques.
- To learn Object oriented programming.
- To use JAVA application development platforms and Android application development platform.

Course Outcomes

- Students learn OOP development on Eclipse and developer platform.
- Students learn GUI based programming.
- Students learn to develop android and mobile applications.
- Students learn to write larger complex applications.

Syllabus

Introduction to OOPs and Java Language Fundamentals - OOPs Principles, Features of Java. JVM, Bytecode, JRE, Language Fundamentals.

Classes and Objects- Introducing class fundamentals, Object and Object reference, Introducing methods, Extending Objects, Object Life time & Garbage Collection, Creating and Operating Objects, Constructor & initialization code block, Access Control, Modifiers, Nested, Inner Class and Anonymous Classes, Abstract Class and Interfaces, Defining Methods, Argument Passing Mechanism, Method Overloading, Recursion, Dealing with Static Members, finalize method, native Method. Use of “this” reference, Use of Modifiers with Classes & Methods, Method overriding, Package and Interfaces, Garbage Collection. Extending Classes and Inheritance: Use and Benefits of Inheritance in OOP, Types of Inheritance in Java, Role of Constructors in inheritance, Overriding Super Class Methods, use of “super”, Polymorphism in inheritance, Implementing interfaces.

Package Organizing Classes and Interfaces in Packages, Package as Access Protection.

Exception Handling: Exceptions & Errors, Types of Exception, Control Flow in Exceptions.

Array & Strings: Defining an Array, Initializing & Accessing Array, Multi –Dimensional Array, Operation on String, Mutable & Immutable String, Using Collection Bases Loop for String, Tokenizing a String, Creating Strings using String Buffer.

Application problems on Linked list, stack in Java, Priority Queue in Java using comparators, hashmap, File Input/Output.

Introduction to PHP app development- Introduction to PHP, Basic Syntax of PHP, PHP statement terminator and case insensitivity, Embedding PHP in HTML, Comments, Variables, Assigning value to a variable, Constants, Managing Variables; Operators and Controls Structures; Functions in PHP, Arrays, PHP File and Forms Handling- File Open, File Creation, Writing to files, Reading from File, Searching a record from a file, Closing a File, PHP Server-side programming - Using PHP With HTML Forms, GET and POST methods, Sessions and Cookies, Support for Database, Creating classes in PHP.

Textbooks/ References

1. C. Thomas Wu, An Introduction to Object Oriented Programming with Java, Fifth Edition, 2009.
2. Ken Arnold, James Gosling and David Holmes, The Java Programming Language, Fourth Edition, 2005.
3. Herbert Schildt, Java: The Complete Reference, McGraw Hill Education(India), 11th Edition 2018.
4. 1. Steven Holzner, PHP: The Complete Reference, McGraw Hill Education, 2017.

IHS121 Personality Development [1-0-0-1]

Course Objectives

- To understand the basic perspectives of human personality such as; Trait approach, Psychoanalytic approach, Biological basis, Humanistic/phenomenological approach, Behaviorist/learning, Cognitive approach, Interaction perspective, and Transpersonal perspective (Indian and Yoga Psychology).
- Learn to objectively assess and explain the behavior of other people, identify personality traits so as predict how a person will behave, and to help to function effectively.
- Have understanding how hiring decisions are taken based on personality characteristics that serve as requirements of a job
- Understanding the application of assessment of Type A & B personality on personal health & achievement
- Understand Personality disorders and its identification. Cognitive Behavior Therapy in the context of Psychotherapy for personality disorders.

Course Outcomes

- Understand one's own personality and that of others, appreciate uniqueness of individuals, adapt to people and situations effectively, assess self and others using scientific tools of personality, cope with challenges in life with better understanding of human behavior science.

Syllabus

Personality: Meaning & Assessment. Psychoanalytic & Neo-Psychoanalytic Approach ; Behavioural Approach; Cognitive Approach; Social- Cognitive Approach; Humanistic Approach; The Traits Approach; Models of healthy personality: the notion of the mature person, the self-actualizing personality etc. Personality disorders; Psychotherapeutic techniques and Yoga & Meditation; Indian perspective on personality; Personality in Sociocultural context.

Textbooks/References

1. Schultz, D.P., & Schultz, S. E. (2005)(8th Edn.)Theories of Personality. Belmont: Thomson Wadsworth.
2. Lindzey, G., Campbell, J.B., & Hall, C.S.(2007)(4th Edn.). Theories of Personality. NewYork:Wiley & Sons
3. Ryckman, R.M. (2008)(9th Edn.).Theories of Personality.Belmont: Thomson Wadsworth.
4. Rao, K.R., & Paranjpe, A.C.(2016).Psychology in the Indian Tradition. NewDelhi:Springer.
5. Frankl, V.E.(1992). Man's Search for Meaning. Massachusetts:Beacon Press
6. Simanowitz, V., & Pearce, P. (2003). Personality Development. England: Open University Press.

SEMESTER III

IMA211 Probability, Statistics, and Random Processes [3-1-0-4]

Course Objectives

- To expose the students to the modern theory of probability, concept of random variables and their expectations.
- To introduce various discrete and continuous distributions and concept of estimation theory, confidence interval.
- To illustrate the concept of hypothesis testing, tests for means and variances, Goodness of fit tests
- To introduce the concept of random processes, Markov chains, Brownian Motion

Course Outcomes

- Define and apply the concepts of probability and conditional probability
- Define and illustrate discrete and continuous random variables, their probability mass functions and probability density functions
- Understand the concept and need of hypothesis testing
- Perform the tests for means and variances and Goodness of fit test
- Understand the concept of random processes, Markov chains, Brownian motions

Syllabus

Axiomatic construction of the theory of probability, independence, conditional probability, and basic formulae.

Random variables and distributions: Univariate, Bivariate and multivariate random variables, Cumulative and marginal distribution function, Conditional and multivariate distributions, Functions of random variables: Sum, product, ratio, change of variables.

Mathematical expectations, moments, moment generating function, characteristic functions;Discrete/continuous distributions and limit theorems: Binomial distribution, Geometric distribution, Poisson distribution, Normal distribution, Exponential distribution, Gamma distribution, Beta distribution, Central limit theorem, Tchebeyche's inequality, Law of large numbers.

Estimation Theory: Bias of estimates, Confidence intervals, Minimum variance unbiased estimation, Bayes' estimators, Moment estimators, Maximum likelihood estimators, Chi-square distribution, Confidence intervals for parameters of normal distribution.

Hypothesis testing: Tests for means and variances, hypothesis testing and confidence intervals, Bayes' decision rules, Power of tests, Goodness-of-fit tests, Kolmogorov-Smirnov Goodness-of-fit test.

Definition and classification of random processes, discrete-time Markov chains, Poisson process, continuous-time Markov chains, stationary processes, Gaussian process, Brownian motion.

Textbooks/ References

1. S. Ross, Introduction to Probability and Statistics for and Engineers and Scientists, Third Edition, Elsevier, 2004.
2. P. G. Hoel, S. C. Port and C. J. Stone, Introduction to Probability Theory, Universal Book Stall, 2000.
3. S. M. Ross, Introductory Statistics, Second Edition, Academic Press, 2009.
4. J. Medhi, Stochastic Processes, Third Edition, New Age International, 2009.
5. V.K.Rohati and A.K. Saleh, An introduction to Probability and Statistics, Third Edition. Wiley Student Edition, 2006.
6. G. R. Grimmett and D. R. Stirzaker, Probability and Random Processes, Oxford University Press, 2001.
7. W. Feller, An Introduction to Probability Theory and its Applications, Vol. 1, Third Edition., Wiley, 1968.
8. S.M. Ross, Stochastic Processes, Second Edition. Wiley, 1996.
9. C. M. Grinstead and J. L. Snell, Introduction to Probability, Second Edition, Universities Press India, 2009.
10. S.Ross, A First Course in Probability, 10th Edition, Pearson Education, Delhi, 2018

ICS211 Design and Analysis of Algorithms [3-1-0-4]

Course Objectives

- Analyze the asymptotic performance of algorithms.
- Demonstrate a familiarity with major algorithms and data structures.
- Apply important algorithmic design paradigms and methods of analysis.
- Synthesize efficient algorithms in common engineering design situations.

Course Outcomes

- Argue the correctness of algorithms using inductive proofs and invariants.
- Analyze worst-case running times of algorithms using asymptotic analysis.
- Describe the divide-and-conquer paradigm and explain when an algorithmic design situation calls for it. Recite algorithms that employ this paradigm. Synthesize divide-and-conquer algorithms. Derive and solve recurrences describing the performance of divide-and-conquer algorithms.

- Describe the dynamic-programming paradigm and explain when an algorithmic design situation calls for it. Recite algorithms that employ this paradigm. Synthesize dynamic programming algorithms, and analyze them.
- Describe the greedy paradigm and explain when an algorithmic design situation calls for it. Recite algorithms that employ this paradigm. Synthesize greedy algorithms, and analyze them.
- Explain the major graph algorithms and their analyses. Employ graphs to model engineering problems, when appropriate. Synthesize new graph algorithms and algorithms that employ graph computations as key components, and analyze them.
- Explain what an approximation algorithm is, and the benefit of using approximation algorithms.

Syllabus

Introduction: Efficiency – Run Time & Space. Analyzing an Algorithm – Insertion Sort- Proof of Correctness – Complexity – Asymptotic Notations.

Divide and Conquer: Analyzing Recursive Algorithms – Merge Sort, Recurrence Relations – Binary Search. Solving Divide and Conquer Recurrences – Recursion Tree – Substitution Method – Master Theorem – Applications of the Master Theorem.

Greedy Algorithms: Locally Optimal Solutions – Interval Scheduling – Minimum Spanning Trees.

Prim's Algorithm – Locally Modifying Solutions to Build Better Solutions – Exchange Arguments.

Dijkstra's Algorithm – Kruskal's Algorithm – Knapsack – Huffman Coding.

Dynamic Programming: Reusing work across sub computations – Definition of Dynamic Programming – Optimal Rod Cut Problem - Optimal Matrix Chain Multiplication - Bellman-Ford Algorithm, Floyd-Warshall Algorithm – Longest Common Subsequence – Machine Scheduling Problem. Application problems- Max Flow problems in flow networks.

Amortized Complexity Analysis – Aggregate Method, Accounting Method, Potential Method, Dynamic Tables – Balanced Trees.

Intractable Problems: Polynomial Time – class P – Polynomial Time Verifiable Algorithms – class NP – NP completeness and reducibility – NP Hard Problems – NP completeness proofs – Approximation Algorithms.

Textbooks/ References

1. Thomas H Cormen, Charles E Leiserson, Ronald L Rivest, Clifford Stein - Introduction to Algorithms, MIT Press, Third Edition, 2010.
2. Jon Kleinberg, Eva Tardos, Algorithm Design ,Pearson Addison, Wesley, 2013.

ICS212 Theory of Computation [3-1-0-4]

Course Objectives

- Course will provide a formal connection between algorithmic problem solving and the theory of languages and automata and develop them into a mathematical (and less magical) view towards algorithmic design and in general computation itself.
- The course should in addition clarify the practical view towards the applications of these ideas in the engineering part of CS.

Course Outcomes

- Model, compare and analyse different computational models using combinatorial methods.
- Apply rigorously formal mathematical methods to prove properties of languages, grammars and automata.
- Construct algorithms for different problems and argue formally about correctness on different restricted machine models of computation.
- Identify limitations of some computational models and possible methods of proving them.
- Have an overview of how the theoretical study in this course is applicable to and engineering application like designing the compilers.

Syllabus:

Introduction: Notion of formal language- Strings, Alphabet, Language, Operations, Finite State Machine, definitions, finite automaton model, acceptance of strings and languages, deterministic finite automaton, equivalence between NFA and DFA, Conversion of NFA into DFA, minimization of FSM, equivalence between two FSM's, Moore and Mealy machines.

Regular expressions: Regular sets, regular expressions, identity rules, manipulation of regular expressions, equivalence between RE and FA, inter conversion, Pumping lemma, Closure properties of regular sets, regular grammars, right linear and left linear grammars equivalence between regular linear grammar and FA, inter conversion between RE and RG.

Context free grammars: Derivation, parse trees. Language generated by a CFG. Eliminating useless symbols, productions, and unit productions. Chomsky Normal Form.

Pushdown automata: Definition, instantaneous description as a snapshot of PDA computation, notion of acceptance for PDAs.

Turing machine: Turing machine, definition, model, design of TM, Computable Functions, recursive enumerable language, Church's Hypothesis, Counter machine, types of TM's, RAM machine.

Un-decidability and classes of problems: Chomsky hierarchy of languages, linear bounded automata and context sensitive language, Grammar, decidability of problems, Universal Turing Machine, un-decidability of post's correspondence problem.

Turing reducibility logical theories. Complexity classes: P, NP, co-NP, EXP, PSPACE, L, NL, ATIME, BPP, RP, ZPP, IP.

Textbooks/ References

1. M Sipser, Introduction to the Theory of Computation, Second Edition, Thomson, 2005.
2. Lewis H.P. and Papadimitriou C.H. Elements of Theory of Computation, Prentice Hall of India, Fourth Edition, 2007.
3. S. Arora and B. Barak, Computational Complexity: A Modern Approach, Cambridge University Press, 2009.
4. C. H. Papadimitriou, Computational Complexity, Addison-Wesley Publishing Company, 1994.
5. D. C. Kozen, Theory of Computation, Springer, 2006.
6. D. S. Garey and G. Johnson, Computers and Intractability: A Guide to the Theory of NP- Completeness, Freeman, New York, 1979.
7. J Hopcroft, JD Ullman and R Motwani, Introduction to Automata Theory, Languages and Computation, Third Edition, Pearson, 2008.

ICS213 Database Management Systems [2-1-2-4]

Course Objectives

- To understand the need for a database and its management using DBMS.
- To model Entity-Relationship (ER) diagram for a real-world scenario.
- To write relational algebra and relational calculus queries for data handling and retrieval, Write SQL queries for database creation and analysis.
- Design efficient database systems using the principle of normalization.
- Understand the basics of database transactions, deadlock handling and security.
- How to implement database indexing.
- Usage of tools like RA Interpreter and MySQL for executing various queries.

Course Outcomes

- Understand database concepts and structures and query language.
- Understand the ER model and relational model.
- Apply various Normalization techniques.
- Understand query processing and techniques involved in query optimization.

Syllabus

Database Modeling: Database System concepts and architecture, Data modeling using Entity Relationship (ER) model and Enhanced ER model, Specialization, Generalization.

Database Indexing: Data Storage and indexing- Single level and multi-level indexing, Dynamic Multi level indexing using B Trees and B+ Trees.

Relational Databases: The Relational Model, Relational database design using ER to relational mapping Relational algebra, Relational calculus, Tuple Relational Calculus, Domain Relational Calculus, SQL.

Database Design: Database design theory and methodology, Functional dependencies and normalization of relations, Normal Forms, Properties of relational decomposition, Algorithms for relational database schema design.

Database Transactions: Transaction processing concepts, Schedules and serializability, Concurrency control, Two Phase Locking Techniques, Optimistic Concurrency Control, Database recovery concepts and techniques. Database Security: Introduction to database security.

Textbooks/References

1. Ramez Elmasri and Shamkant B. Navathe, Fundamentals of Database Systems, Fifth Edition, Pearson Education, 2008.

2. Raghu Ramakrishnan and Johannes Gehrke, Database Management Systems, Third Edition, McGraw Hill, 2014.
3. Peter Rob and Carlos Coronel, Database System- Design, Implementation and Management, Seventh Edition, Cengage Learning, 2007.

ICS214 IT Workshop III [2-1-2-4]

Course Objectives

- Learn Python scripting and the scripting shell.
- Master the basics of programming constructs, like conditions, loops, functions, etc.
- Introduce sequence types in Python like Lists, Tuples, Sets and Dictionaries.
- Be exposed to advanced applications such as databases, networks, etc

Course Outcomes

- Write python programs for various applications.
- Write Database programs to create, access, modify and update data.
- Write network programs for sending emails, ftp, sockets etc.

Syllabus

Introduction to data types, variables, constants, operators, input-output, basic formatting, running python programs, date and time functions. Conditionals, if statement and variants, relational operators, logical operators.

Iteration and while loops, for loops and range command, random numbers. File processing, reading and writing files, parsing files, text files and CSV files.

Lists and list processing, list operations, list traversals, tables as lists. Tuples, Maps, Sets and Dictionaries, creation and traversals.

Strings and string processing, string functions, conversions.

User defined Functions, lambda functions, recursive functions, built-in functions, yield statement, parameter passing.

Classes and object-oriented programming, inheritance, associations.

Database processing, creating tables, querying, MySQL and PySqlite.

Network programming, sockets, email sending, ftp Threads and multithreading.

Numpy and applications in matrices, random numbers Scipy, Matplotlib, graphing and charting data.

Trie, stack, Networkx, pattern matching, regular expression.

Introduction to Android app development.

Textbooks/References

1. Ljubomir Perkovic, Introduction to Computing with Python, Wiley, Second Edition, 2015.
2. Narasimha Karumanchi, Data Structures and Algorithms With Python, Careermonk Publications, 2015.

ISC213 Introduction to Cognitive Science [2-0-0-2]

Course Objectives

- To study the basic concepts and approaches in the field of cognitive science
- To apply the concepts of planning, reasoning and learning models in cognitive applications
- To analyse language and semantic models of cognitive process.

Course Outcomes

- Demonstrating the qualitative and quantitative skill and critical thinking on cognitive science by applying suitable methodology to real world cognitive applications
- Acquire knowledge in language processing and understanding
- Learning the cognitive processes of humans, robots and other intelligent systems

Syllabus

Introduction to Cognitive Science: Fundamental Concepts of cognitive science - Computers in Cognitive Science - Applied Cognitive Science - The Interdisciplinary Nature of Cognitive Science - Structure and constituents of the brain; Brief history of neuroscience; Mathematical models; Looking at brain signals; Processing of sensory information in the brain.

Memory: Types of Memory – Memory Models - Visual Imagery - Problem Solving - Overall Evaluation of the Cognitive Approach.

Network Approach: Principles Underlying Artificial Neural Networks (ANN) - Characteristics of ANN – Conceptions of Neural Networks - Back Propagation and Convergent Dynamics - ANN Typologies - Evaluating the Connectionist Approach - Semantic Networks - Characteristics of Semantic Networks - Evaluation of the network approach.

Cognitive Science in Linguistics: Importance of Language – Nature Language - Language Use in Primates - Language Acquisition - Language Deprivation - Cognition and Linguistics: The Role of Grammar - Neuroscience and Linguistics - Artificial Intelligence and Linguistics – Speech Recognition - Evaluation of Natural Language Processing.

Cognitive Science in Robotics: Affordances, direct perception, Ecological Psychology, affordance learning in robotics; Development, child and robotic development; Attention and related concepts; Human visual attention; Computational models of attention; Applications of computational models of attentional.

Textbooks/References

1. Bermúdez, José Luis. Cognitive science: An introduction to the science of the mind. Cambridge University Press, 2014
2. Neil Stillings, Steven E. Weisler, Christopher H. Chase and Mark H. Feinstein, Cognitive Science: An Introduction, Second Edition, MIT press ,1995
3. Mallick, Pradeep Kumar, Borah, Samarjeet, Emerging Trends and Applications in Cognitive computing, IGI Publishers 2019
4. Fromkin, Rodman, and Hyams. An Introduction to Language, Boston, MA: Thomson Wadsworth, 9th edition, 2011
5. Stuart J. Russell, Peter Norvig, Artificial Intelligence - A Modern Approach, Third Edition, Pearson Publishers, 2015.

ICS215 Data Structures II [1-0-2-2]

Course Objectives

- Teach programming with emphasis on problem solving and introduce Data structures.
- Provide the Foundations of the practical implementation and usage of Algorithms and Data Structures.

Course Outcomes

- Design correct programs to solve problems.
- Choose efficient data structures and apply them to solve problems.

Syllabus

Problems on Geometric algorithms but not limited to: Klee's Algorithm, Manhattan distance problems, Collinear checking, Identifying Integral points inside a Triangle, Circumcenter of a Triangle, Triangular Matchstick Number, area of Circumcircle of an Equilateral Triangle, Number of rectangles in $N * M$ grid, Area of two overlapping rectangles, Number of unique rectangles formed using N unit squares, Circle and Lattice Points, Pizza cut problem.

Algorithms on Bit Manipulation but not limited to: Letter manipulation problems, k-th bit Manipulation problems, Kernighan's Algorithm to count set bits in an integer.

Discussion on Numerical algorithms but not limited to: Gauss-Jordan Elimination, Matrix Manipulation problems.

Textbooks/References

1. Cormen, Thomas H., et al. Introduction to algorithms. MIT press, 2009.
2. Aho, Albred V., J. E. Hopcroft, and J. D. Ullman. "Data structures and algorithms (1983).
3. Drozdek, Adam. Data Structures and algorithms in C^{++} . Cengage Learning, 2012.
4. Allen, Weiss Mark. Data structures and algorithm analysis in C^{++} . Pearson Education India, 2007.
5. Kleinberg, Jon, and Eva Tardos. Algorithm design. Pearson Education India, 2006.
6. Skiena, Steven S. The algorithm design manual. Springer International Publishing, 2020.
7. Knuth, Donald Ervin. The art of computer programming. Vol. 3. Pearson Education, 1997.
8. Nomura, Seiichi. "C Programming and Numerical Analysis: An Introduction." Synthesis Lectures on Mechanical Engineering 2.2 (2018): 1-198.
9. Dasgupta, Sanjoy, C. H. Papadimitriou, and U. V. Vazirani. "Algorithms; 2006."
10. Trefethen, Lloyd N., and David Bau III. Numerical linear algebra. Vol. 50. Siam, 1997.

SEMESTER IV

IMA221 Differential Equations and Transforms [3-1-0-4]

Course Objectives

- Find the Fourier series representation of a function of one variable
- Introduce the Fourier series and its application to the solution of partial differential equations
- Introduce the concepts of Laplace and Fourier transforms.
- Identify the type of a given differential equation and select and apply the appropriate analytical technique for finding the solution of first order and selected higher order ordinary differential equations.
- Introduce students to partial differential equations
- Introduce students to how to solve linear Partial Differential with different methods.

Course Outcomes

- Analyse and solve engineering problems using Fourier series.
- Find the Laplace and Fourier transforms of functions of one variable.
- Solve first order differential equations utilizing the standard techniques for separable, exact, linear, homogeneous, or Bernoulli cases. Find particular solutions when given initial or boundary conditions.
- Will be able to find solution of higher-order linear differential equations
- Classify PDEs, apply analytical methods, and physically interpret the solutions.

Syllabus

Fourier Series : Dirichlet's conditions - General Fourier series - Odd and even functions - Half range Sine and Cosine series - Complex form of Fourier series - Parseval's identity - Harmonic Analysis. Convergence of FS, differentiation and integration of Fourier series.

Fourier Transform Fourier Integral Theorem Fourier transform pair - Sine and cosine transforms - Properties - Transform of elementary functions - Convolution theorem - Parseval's identity.

Ordinary Differential Equations: Method of variation of parameters - Method of undetermined coefficients - Homogeneous equation of Euler's and Legendre's type - System of simultaneous linear differential equations with constant coefficients.

Partial Differential Equations Formation Solutions of first order equations - Standard types and Equations reducible to standard types Singular solutions - Lagrange's linear

equation Integral surface passing through a given curve Classification of partial differential equations Solution of linear equations of higher order with constant coefficients - Linear non-homogeneous.

Fourier Series Solutions of Partial Differential Equations: Method of separation of variables. Solutions of one dimensional wave equation and one dimensional heat equation - Steady state solution of two-dimensional heat equation Fourier series solutions in Cartesian coordinates.

Textbooks/References

1. C. Edwards and D. Penney, Elementary Differential Equations with Boundary Value Problems, 6th edition, Pearson, 2003
2. W.E. Boyce and R.C. DiPrima, Elementary Differential Equations, 7th Ed., John Wiley & Sons, 2002.
3. Erwin Kreyszig, Advanced Engineering Mathematics, 10th Edition, Wiley, 2015.
4. Tyn Myint-U, L. Debnath, Linear Partial Differential Equations for Scientists and Engineers, 4th Edition, Birkhauser, 2007.

ICS221 Operating Systems [3-1-0-4]

Course Objectives

- To introduce the Fundamental concept of OS, and how OS works.
- To study the Building blocks of OS, Components of OS.

Course Outcomes

- Students will be able to design an OS.
- Students will be able to implement various components of OS.
- Students will be able to implement a small OS.

Syllabus

Operating system overview: Computer System Organization, Operating System structure, operations of OS, process management, memory management, storage management, protection and security, distributed systems.

Processes: Process concept, Process scheduling, Operations on processes, Cooperating processes, inter-process communication.

Threads: Overview, Multi-threading models, threading issues, P threads, Windows XP threads.

CPU Scheduling: Basic concepts, scheduling criteria, scheduling algorithms, multiple-processor scheduling.

Process synchronization: The critical section problem, Peterson's solution, synchronization hardware, Semaphores, Monitors, Synchronization examples. Deadlocks: Methods for handling deadlocks, Deadlock prevention, deadlock avoidance, Deadlock recovery.

Memory management: Swapping, Paging, Segmentation, Virtual memory, Demand paging, Page replacement.

I/O Systems: I/O hardware, Application I/O interface, Kernel I/O subsystem, transforming I/O requests to hardware operations.

Textbooks/References

1. William Stallings, Operating systems: Internals & design principles, Pearson, Seventh edition, 2014.
2. Andrew S. Tanenbaum, Modern Operating Systems, Pearson Fourth Edition, 2016.
3. Charles Crowley, Operating Systems - Design Oriented Approach, Mc. Graw Hill Education, First edition, 2017.
4. Abraham Silberschatz, Galvin, Gagne Operating System Concepts, Wiley, Ninth Edition, 2016.
5. J. Millman & C.C. Halkias-Integrated Electronics, 2nd edition, 2010, TMH. ISBN 0- 07462245-5.

ICS222 Object-Oriented Analysis and Design [3-0-2-4]

Course Objectives

- To understand the Object-based view of Systems.
- To develop robust object-based models for Systems.
- To inculcate necessary skills to handle complexity in software design.
- To understand importance of Architectural patterns and Design patterns.

Course Outcomes

- Ability to analyze and model software specifications.
- Ability to abstract object-based views for generic software systems.
- Ability to deliver robust software components using design patterns.

Syllabus:

Introduction to Systems Concepts: Definition. Characteristics of a System: Organization, Interaction, Interdependence, Integration. Elements of a System: Inputs and Outputs, Processor(s), Control, Feedback, Environment. Types of Systems: Physical/Abstract Systems, Open/Closed Systems, Man-made Information Systems. System Modelling: The importance of Modelling, Principles of Modelling, Object-Oriented Modelling, Introduction to UML. Best Practices of Software Engineering: Develop Iteratively, Models and visualization, UML, Software Development Life Cycle. Rational Unified Process: Process, Phases in RUP, RUP Workflows Process, Lifecycle. Design Pattern, Architectural Design Patterns, Describing Design Patterns, The Catalog of Design patterns, How to select a Design Pattern, How to use a Design Pattern.

Concepts of Object Orientation: Object Modelling Technology, Basic Principles of Object Orientation: Abstraction Encapsulation, Polymorphism, Modularity, Hierarchy. UML Modeling Mechanisms. Structural Modelling: Classes and Interfaces, Attributes and Operations, Relationships: Association, Dependency, Generalization, Multiplicity, Aggregation, Interface, Types and Roles, Instances. UML Notation: Things, Relationships, Extensibility, Diagrams. UML Standard Elements: Stereotypes, Tagged Values, Constraints.

Diagrams: Structural Diagrams and Behavioral Diagrams -Class Diagrams, Object Diagrams. Designing an object-oriented system: Identifying the classes, assigning attributes and behaviour, finding relationship between classes, arranging classes into hierarchies: A design example. Requirement Overview: Software Requirement Specification, Use case Model, Use case Diagrams. Analysis and Design: Use case Analysis, Use case behaviour, Finding Classes, Class Analysis, Attributes and Associations.

Interaction Diagrams: Sequence Diagrams, Collaboration Diagrams, Activity Diagrams: Activity States, Transitions, State Chart Diagram. Component Diagrams, Deployment Diagrams: Distribution Diagrams, Runtime Architecture, Concurrency, Process, Nodes, Networks, Deployment Model.

Architectural Patterns – Layered architecture, Pipers and Filters, Blackboard, Broker, MVC, MVVM, Micro-Kernel, Master-Slave, PAC, others.

Design Pattern, Architectural Design Patterns, Describing Design Patterns, The Catalog of Design patterns, How to select a Design Pattern, How to use a Design Pattern. Creational Patterns : Singleton, Abstract Factory, Builder, Factory Method, Prototype.

Structural Pattern: Adaptor, Bridge, and Composite, Decorator, Arcade, Flyweight, Proxy.

Behavioral Patterns: Chain of Responsibility, Command, Interpreter, and Iterator, Mediator, Memento, Observer, State, Strategy, Template Method, Visitor.

Textbooks/ References

1. Grady Booch James Rumbaugh, Ivar Jacobson, The Unified Modeling Language User Guide, Addison Wesley, 2012.
2. Grady Booch, Object Oriented Analysis and Design With Applications, Third Edition, Pearson Education, 2009.
3. Michael Blaha, James Rumbaugh, Object Oriented Modeling and Design with UML, Second Edition, Pearson Education, 2005.
4. Martin Fowler, UML Distilled: A Brief Guide to Standard Object Modeling, Third Edition, Addison Wesley, 2015.
5. Brett D. McLaughlin, Gary Pollice, and Dave West, Head First Object-Oriented Analysis and Design, First Edition,

O'Reilly, 2006.

6. Elias M Awad, System Analysis and Design, Second Edition, Galgottia Publications Pvt. Ltd, 2015.
7. Eric Freeman, Elisabeth Freeman, Kathy Sierra, Bert Bates Head First Design Patterns, O'Reilly
8. Erich Gamma, Richard Helm, Ralph Johnson, John Vlissides Design Patterns: Elements of Reusable Object Oriented Software Addison-Wesley
9. Pattern Oriented Software Architecture, af.Buschman & others, John Wiley & Sons.
10. Design Patterns Explained By Alan Shallowy, Pearson Education.

ICS223 Compiler Design [3-0-2-4]

Course Objectives

- To make the students capable of applying the principles, algorithm, and data structures involved in the design of compilers.
- Students should be able to design a lexical analyser in lex according to the specification. They should be able to design a parser in yacc when the specification is mentioned.
- They should be able to construct a compiler according to the rules and constraints given.
- To introduce the major concept areas in compiler design and know the various phases of the compiler
- To understand the various parsing algorithms and comparison of the same
- To provide practical programming skills necessary for designing a compiler
- To gain knowledge about the various code generation principles
- To understand the necessity for code optimization

Course Outcomes

- To introduce the major concept areas of language translation and compiler design.
- To enrich the knowledge in various phases of compiler and the design issues involved in compilation, code optimization techniques, machine code generation, and use of symbol table.
- To extend the knowledge of parser by parsing LL parser and LR parser.
- To provide practical programming skills necessary for constructing a compiler.
- Apply the knowledge of LEX, YACC, ANTLR tools to develop a scanner and parser
- Suggest the necessity for appropriate code optimization techniques

- Conclude the appropriate code generator algorithm for a given source language
- Design a compiler for any programming language

Syllabus

Introduction to programming language translation. Lexical analysis: Specification and recognition of tokens.

Syntax analysis: Top-down parsing-Recursive descent and Predictive Parsers. Bottom-up ParsingLR (0), SLR, and LR (1) Parsers.

Semantic analysis: Type expression, type systems, symbol tables and type checking. Intermediate code generation: Intermediate languages. Intermediate representation-Three address code and quadruples. Syntax-directed translation of declarations, assignments statements, conditional constructs and looping constructs.

Runtime Environments: Storage organization, activation records. Introduction to machine code generation and code optimizations.

Lab Practice

- Generation of lexical analyzer using tools such as LEX - Generation of parser using tools such as YACC - Creation of Abstract Syntax Tree- Creation of Symbol tables, Semantic Analysis - Generation of target code.

Textbooks/References

1. Aho A.V., Lam M. S., Sethi R., and Ullman J. D., Compilers: Principles, Techniques and Tools, Pearson Education, 2007.
2. Appel A.W, and Palsberg J., Modern Compiler Implementation in Java, Cambridge University Press, 2002.
3. W. Appel, Modern Compiler Implementation in C, Cambridge University Press, 1998.
4. V. Aho, M. S. Lam, R. Sethi, J. D. Ullman, Compilers-Principles, Techniques & Tools, Second Edition, Pearson Education, 2007.

ICS224 Computer Networks [3-0-2-4]

Course Objectives

- The students should understand the layers of networking devices.
- They should be familiar with a few networking protocols.
- They should study the different types of networks and topologies of networks.

Course Outcomes

- To distinguish the importance of different networking components.
- To understand the functionalities of each networking layers and standards.

- To write simple networking-based programs at real and simulator level.

Syllabus

Evolution of computer networks: Network Architecture-OSI, TCP/IP models. Physical and Data link layer: Encoding, Framing, Error detection, HDLC, PPP, sliding window protocols, medium access control, Token Ring, Wireless LAN, Packet Switching.

Network Layer: Internet addressing, IP, ARP, ICMP, CIDR, Routing algorithms (RIP, OSPF, BGP).

Transport Layer: UDP, TCP, flow control, congestion control, Introduction to quality of service.

Application Layer: DNS, Web, HTTP, email, authentication, encryption.

Lab practice

- Unix network measurement and analysis tools, NS3 Socket interface and programming, RPC, RMI, Assignments using Network Simulators.

Textbooks/References

1. L. L. Peterson and B. S. Davie, Computer Networks: A Systems Approach, Fifth Edition, Elsevier, 2011.
2. A. S.Tanenbaum and D.J. Wetherall, Computer Networks, Fifth Edition, Pearson, 2011.
3. W. R. Stevens, UNIX Network Programming, Volume 1: Networking APIs: Sockets and XTI, Second Edition, PrenticeHall,1998.
4. S. S. Panwar, S. Mao, J. Ryoo, and Y. Li, TCP/IP Essentials: A Lab-based Approach, Cambridge Press, 2004.
5. J. F. Kurose and K. W. Ross, Computer Networking: A Top Down Approach, Seventh Edition, Pearson India, 2017.
6. D. E. Comer, Internetworking with TCP/IP Vol. 1, Sixth Edition, Prentice Hall of India, 2006.
7. B. Forouzan, Data Communications and Networking, Fifth Edition, Tata Mcgraw Hill, 2012.
8. Introduction to Network Simulator NS2, Second Edition, 2011.

ICS225 Data Structures III [1-0-2-2]

Course Objectives

- Ensure that the student evolves into a competent programmer capable of designing and analyzing implementations of algorithms and data structures for different kinds of problems.
- Expose the student to the algorithm analysis techniques.

Course Outcomes

- Analyze the efficiency of programs based on Complexity.

- Understand the necessary mathematical abstraction to solve problems.

Syllabus

Algorithms on Graph connectivity but not limited to: Tarjan's and Kosaraju's strongly connected components algorithms, Detect cycles in an undirected graph, Degree of vertices in a Graph, Path identification between vertices in Undirected graph.

Discussions on Randomized algorithms but not limited to: Reservoir Sampling, Birthday Paradox, Load Balancing problem, Karger's algorithm for Minimum Cut, Freivald's Algorithm to check the product of a matrix, Monte Carlo estimation.

Branch and Bound Algorithms: Knapsack problem, Travelling salesman problem.

Threaded Binary Tree, Splay trees, Foldable binary trees, Additional problems on BST, Binomial heap, Fibonacci heap, Topological sorting, self-organizing tree, segment tree, Binary indexed tree, suffix array and suffix tree, pattern searching, Tribonacci word.

Textbooks/References

1. Cormen, Thomas H., et al. Introduction to algorithms. MIT press, 2009.
2. Aho, Albrecht V., J. E. Hopcroft, and J. D. Ullman. "Data structures and algorithms (1983).
3. Drozdek, Adam. Data Structures and algorithms in C++. Cengage Learning, 2012.
4. Allen, Weiss Mark. Data structures and algorithm analysis in C++. Pearson Education India, 2007.
5. Kleinberg, Jon, and Eva Tardos. Algorithm design. Pearson Education India, 2006.
6. Skiena, Steven S. The algorithm design manual. Springer International Publishing, 2020.
7. Knuth, Donald Ervin. The art of computer programming. Vol. 3. Pearson Education, 1997.
8. Nomura, Seiichi. "C Programming and Numerical Analysis: An Introduction." Synthesis Lectures on Mechanical Engineering 2.2 (2018): 1-198.
9. Dasgupta, Sanjoy, C. H. Papadimitriou, and U. V. Vazirani. "Algorithms; 2006
10. Trefethen, Lloyd N., and David Bau III. Numerical linear algebra. Vol. 50. Siam, 1997.

IHS221 Fundamentals of Economics [1-0-0-1]

Course Objectives

- To familiarize the participants concepts and techniques in Economics
- To make the participants appreciate the applications of core concepts in economics for managerial decision making

- To sensitize the participants how economic environment affects Organizations

Course Outcomes

- It will help the students to analyze the demand and supply conditions and assess the positions of a company.
- It will help to design competition strategies, including costing, pricing, product differentiation and market environment according to the natures of products and structures of market

Syllabus

Introduction to Fundamentals of Economics: Micro & Macro Economics, Managerial Economics – Definition – Nature & Scope.

Fundamental concepts in Managerial economics for decision making: Incremental Principle, Opportunity Cost, Discounting Principle, Time Concept, Equi-Marginal Principle – Illustrations, Decision Making – Process and Conditions – Difference between Risk & Uncertainty.

Demand Analysis and Forecasting: Meaning of Demand – Types of Demand – Law of Demand & its Exceptions, Elasticity of Demand – Price Elasticity, Income Elasticity, Cross Elasticity, Promotion Elasticity, Applications of the concepts of Elasticity, Demand Forecasting – Process – Statistical & Non-Statistical Techniques, Utility Analysis & Consumer Behaviour – Equilibrium of the consumer using Cardinal & Ordinal Utility (Indifference Curve) Theories.

Supply & Production: Theory of Production – Meaning of Production function, Production function with one variable input – Law of Variable Proportions – Returns to Scale, Production function with two variable inputs – Iso-quants – Producers' Equilibrium, Economies of Scale – Types – Economies of Scope, Theory of Costs – Classification of Costs – Short Run & Long Run Cost Curves, Revenue Curves.

Market Structure: Market – Meaning & Elements, Classification of Markets – Markets based on Competition, Theory of Firm – Profit Maximization Rules, Price & Output Determination under Perfect Competition, Price & Output Determination under Monopoly – Monopoly Price Discrimination, Price & Output Determination under Monopolistic Competition, Price & Output Determination under Oligopoly – Kinked Demand curve model only.

Macro Economic Concepts: National Income Concepts – Measurement of National Income, An overview of Financial System in India, An overview of Fiscal & Monetary Policies in India, Balance of Payments: Causes of Disequilibrium & Remedies, Inflation in India – Causes & Remedies. Free Market Economy & Need for Government Intervention – An appraisal of Economic Reforms in India

Textbooks/References

1. Dwivedi D.N, Managerial Economics, Vikas Publications (ISBN 8125910042)

2. P.L. Mehta, Managerial Economics Analysis, Problems and Cases – Sultan Chand & Sons (ISBN 81-7014-386-1)
3. K.K. Dewett, Modern Economic Theory: Micro & Macro Analysis – Orient Book Distributors, New Delhi.
4. V.L. Mote, Managerial Economics – Tata McGraw Hill, New Delhi
5. Gaurav Dutt & Aswani Mahajan, Dutt & Sundaram's Indian Economy – Sultan Chand & Sons.

IHS222 Principles of Management [1-0-0-1]

Course Objectives

- To introduce Profession of Management and help the students gain understanding of the functions and responsibilities of the manager.
- To provide participants tools and techniques to be used in the performance of the managerial job, and enable them to analyze and understand the environment of the organisation

Course Outcomes

- It will help the students to gain understanding of the functions and responsibilities of managers
- It will provide them tools and techniques to be used in the performance of managerial job

Syllabus

Management - Meaning, Definition and Nature; Evolution of Management - Management Thoughts - Early - Modern - Post modern; Contributions of F.W. Taylor - Henry Fayol Hawthorne Studies-Behavioural School of Management Approach. Levels of Management, Skills required for a manager, Managerial roles. Management Lessons from Indian Philosophy Vision, Effectiveness, Efficiency and Teamwork.

Functions of Management: POSDCORB; Characteristics of Management;

Planning: -Meaning - nature - importance -Levels of planning. Objectives - setting objectives Policies - Planning premises, Types of plans - Process of planning - Decision Making; MBO; Principles in Planning.

Organising: - Nature-Purpose-Principles Organisational Structure and types Departmentation - Centralization vs. Decentralization - Span of control- Delegation of Authority - Principles in Organising - Line Vs Staff Authority - Networking and Virtual Organizations Staffing: - Meaning, Principles in Staffing, Staffing Functions.

Directing: - Leadership - Leadership Traits Leadership Styles - Principles in Directing Emerging Trends in Management; Management of Creativity & Innovation - Creative Process Managing E-Business World - Challenges Management in Globalized Era - Organizational Social Responsibility Control:-System and process of Controlling - Requirements for effective control - The Budget as Control Technique - Information

Technology in Controlling - Control Techniques- Control and planning- Types of Control- Reporting - Coordination; Principles in Control and Coordination.

Textbooks/References

1. K.Aswathapa, " Essential of Business Administration", Himalaya Publishing House
2. Harold Koontz & Heinz Weihrich, "Essentials of Management", Tata McGraw-Hill,1998
3. JAF Stomer, Freeman R. E and Daniel R Gilbert, "Management", Pearson Education, Sixth Edition, 2004.
4. Stephen P. Robbins and Mary Coulter, "Management", Prentice Hall of India, 8th edition.
5. Tripathy PC and Reddy PN, "Principles of Management", Tata McGraw-Hill, 1999.
6. Y.K. Bhushan, "Fundamentals of Business Organisation & Management"; Sultan Chand & Co., New Delhi

SEMESTER V

CSE311 Artificial Intelligence [3-0-2-4]

Course Objectives

- To introduce the student about the principles of AI & techniques and do exercises in the laboratory to increase the subject understanding.

Course Outcomes

- Acquire a thorough knowledge and fundamental concepts and techniques of artificial Intelligence.
- Learn simulating tools and study AI language for problem solving
- To develop and test mini intelligent systems.

Syllabus

Introduction to Artificial Intelligence: Artificial Intelligence (AI), Major Branches of AI, Applications- Characteristics and Fundamental issues for AI problems, Steps to build Artificial intelligence (AI) systems, Intelligent systems, Characteristics of intelligent systems.

Search Techniques: Why Search, Applications of search, Tree and Graph, Search strategies, Complexity of Search.

Knowledge Representation: Knowledge, Characteristics of knowledge representation, Types of knowledge representation, Propositional Logic, Tautology and Contradiction, Predicate Logic, Production Systems, Semantic network, Frame systems, Scripts.

Neural Networks: Introduction to Neural network, Structure of Neural network, Structure of Neural network, Neural Network Architecture, Network Layers, Neural Network Learning, Back-propagation Algorithm.

Intelligent agents: Introduction to Agents, Functions, Examples of Agents, Intelligent Agent classification, Features of intelligent agents, Structure of Agents, Intelligent Agents Models Fuzzy logic: Crisp logic, Fuzzy logic, Member ship function, Member ship function, Fuzzy logic Applications.

Expert Systems: What is Expert system, Conventional systems vs. Expert systems, Basic Concepts, Human Expert Behaviors, Knowledge Types, Inferencing, Rules, Structure of Expert Systems, ES Components, Knowledge Engineer, Expert Systems Working, Problem Areas Addressed by Expert Systems, benefits/limitations- Applications of expert systems.

Textbooks/References:

1. Stuart J Russell, Peter Norving. Artificial Intelligence: A Modern approach, Third Edition, 2015.
2. Elaine Rich and Kevin Knight, Introduction to Artificial Intelligence, McGraw Hill, Third Edition, 2017.
3. Michael Negnevitsky, Artificial Intelligence: A guide to Intelligent Systems, Addison Wesley, Third Edition, 2017.

4. G.F. Luger, and W.A. Stubblefield, Artificial Intelligence: Structures and Strategies for Complex Problem Solving, Addison-Wesley Publishing Company, 2011

5. C.S. Krishnamoorthy and S. Rajeev, Artificial Intelligence and Expert Systems for Engineers by CRC Press, 1996.

CSE312 Software Architecture: Principles and Practices [3-0-2-4]

Course Objectives

- The objective of this course is to prepare the students for successful careers in software engineering or related domain through a thorough study of software engineering principles.

Course Outcomes

- Acquire a thorough knowledge of the fundamental concepts in software engineering.
- Apply Learn the actual need for software engineering and various software development life cycle models which will be helpful in software development process.
- Analyse Study various approaches in software architecture and design for effective product development and the major techniques in system testing for better software development process.
- Apply various testing strategies using industrial standard tools and frameworks

Syllabus

Introduction to software engineering: Scope and necessity of software engineering-Evolution of software design techniques-Recent challenges in software industry.

Software life cycle model: Need for software life cycle model-Different life cycle models-Waterfall model-Iterative waterfall model-Prototyping model-Evolutionary model-Spiral model-Agile development methodologies-Rational unified process (RUP)- Extreme Programming (XP).

Requirement analysis and specification: Requirements engineering-Types of system requirements-Role of system analyst-Software requirement specification, -Formal requirement specification IEEE standard.

Architecture and Design: Architectural Styles – Layered architecture, Pipes and Filters, Blackboard, Broker, MVC, MVVM, Micro-Kernel, Micro-Service, Master-Slave, PAC. Design concepts, Coupling and cohesion, Design methodologies-Function Oriented design, UML and Object Oriented design System Design: System modeling- Unified modeling language (UML)-Design Challenges-Design Practices- Top-down and bottom-up design- Experimental prototyping- Collaborative design.

Basic concepts in user interface design: Characteristics of a user interface-Types of user interfaces-Component based graphical user interface design.

Software testing: Verification and validation, Testing concepts- Failure, fault, test case, test suite and test script. Levels of testing, Test plan, Test metrics and coverage, Role of testing-Testing strategies- -Black box and white box testing, Unit tests-Integration testing-Top down integration-Bottom up integration-Validation testing-Alpha testing-Beta testing- Other forms of high-level testing-Stress testing-Code inspections-Manual testing-Automated testing-Breaking tests-Regression testing-Test case execution using testing frameworks Examples of testing frame works (Tinderbox, JUnit, PyUnit) and test automation tools(Selenium, Cucumber).

Software Project Management: Cost estimation- Project scheduling- Staffing-Software configuration management-Quality assurance-Software quality models-Project Monitoring-Risk management, etc.

Emerging Trends in Software Engineering: Observing software engineering trends-Identifying “Soft trends”-Technology directions-Tools related trends.

Text Book/References

1. Roger S Pressman, Software Engineering: A Practitioner’s Approach, McGraw-Hill Higher Education, 7th Edition.
2. Ian Sommerville, Software Engineering, Pearson Education, 9th Edition.
3. Paul, C., & Jorgensen, D. (2021). Software Testing: A Craftsman’s Approach. Auerbach Publishers, Incorporated.
4. Mark; Ford Richards Neal. (2020). Fundamentals of software architecture: an engineering approach. O’Reilly

CSS311 Parallel and Distributed Computing [3-0-2-4]

Course Prerequisites

- Computer Networks
- Fundamentals of Programming
- Computer Organization

Course Objectives

- To introduce the concept and the basics of parallel and distributed computing,
- To provide knowledge on various available parallel programming models
- To enable awareness on the utilization of programming models on various scientific applications

Course Outcomes

- The students will have knowledge on various programming languages for HPC applications.
- The students will have sufficient practical knowledge to utilize the performance analysis tools.

- The students will know the applications of parallel and distributed computing models.

Syllabus

Architectures – Multi-core and Many-core architectures, Accelerators (SIMD units -Vectorization, GPUs), Goals of parallel systems.

Applications – Scientific applications, Characteristics, requirements, regular grid applications, irregular applications, data dependence, parallelization process.

Parallel Programming on Shared Memory – OpenMP (C++ languages), Execution Model, Shared and private data, Directives, Barriers, Sections, Run-Time library functions, scheduling strategies, Scalability study, OpenMP for accelerator programming.

Parallel Programming on Distributed Memory – MPI, Collective operations, Non-Blocking Collectives, Process topologies, Parallel I/O, Single sided communications.

Performance Tools – Concepts, Event-Model execution, profiling, tracing, types of profiling, profiling tools – Scalasca, Score-P, MPI-P, Energy Analyzer, Tracing tools, Autotuning – Periscope Tuning Framework.

Textbooks/References

1. Ian Foster: Designing and Building Parallel Programs – Concepts and tools for Parallel Software Engineering, Pearson Publisher, 1st Edition, 2019.
2. Using OpenMP – The Next Step Using OpenMP – The Next Step – by Ruud van der Pas, Eric Stotzer and Christian Terboven (2017)
3. Parallel Programming in C with MPI and OpenMP by Michael J.Quinn
4. The International Journal of Parallel Programming – Issues and articles devoted to OpenMP, MPI, CUDA.

CSSXXX Cyber Physical Systems and Internet of Things [3-0-2-4]

Course Prerequisite

- Computer Networks.

Course Objectives

- To develop an exposition of the challenges in implementing a cyber-physical system.
- To expose to real world problems on Cyber Physical Systems and provide a walk through the design and validation problems.
- To understand the theory and practice of data sensing and manipulation, engineering foundation of the cyber-physical interactions.

Course Outcomes

- Understand the concept of Internet of Things

- Implement interfacing of various sensors with Arduino/Raspberry Pi.
- Demonstrate the ability to transmit data wirelessly between different devices.

Syllabus:

Introduction: Introduction to IOT, Sensing, Actuation, Cyber-Physical Systems (CPS) in the real world, Communication, Networking.

Industry 4.0: LEAN Production Systems; Cyber Physical Systems and Next Generation Sensors, Collaborative Platform and Product Lifecycle Management;

Basics of Industrial IoT (IIoT): Industrial Processes, Industrial Sensing & Actuation IIoT Communication, IIoT Networking.

Industrial IoT (IIoT) Business Model and Reference Architecture: Business Models, IIoT Reference Architecture; Industrial IoT- Layers: IIoT Sensing, IIoT Processing, IIoT Communication.

IIoT Analytics: IIoT Analytics - Introduction, Machine Learning and Data Science; SDN in IIoT, Data Center Networks, Industrial IoT.

Industrial IoT: Security and Fog Computing - Fog Computing in IIoT, Security in IIoT, Industrial IoT- Application Domains.

Industrial IoT- Application Domains: Healthcare, Power Plants, Inventory Management & Quality Control, Plant Safety and Security (Including AR and VR safety applications), Facility Management; Oil, chemical and pharmaceutical industry, Applications of UAVs in Industries, Real case studies.

Lab Practice

- Introduction to development boards: Arduino, Zigbee, GSM module, Bluetooth module.
- Introduction to Raspberry PI platform and python programming. Interfacing sensors to Raspberry PI, Communicate between Arduino and Raspberry PI using any wireless medium. Setup a cloud platform to log the data, Log Data using Raspberry PI and upload to the cloud platform. Design an IoT based system, System modeling, Control design, stability, Z3 solver.

Textbooks/References

1. Misra, Sudip, Chandana Roy, and Anandarup Mukherjee. Introduction to industrial Internet of Things and industry 4.0. CRC Press, 2021.
2. Alur, Rajeev. Principles of cyber-physical systems. MIT press, 2015.
3. Lee, Edward Ashford, and Sanjit Arunkumar Seshia. Introduction to embedded systems: A cyber-physical systems approach. Mit Press, 2016.

IEG311 Digital Signal Processing [3-0-2-4]

Course Objectives

- To understand the key theoretical principles underlying DSP.
- To analyze and implement digital signal processing systems in time domain.
- To sample and reconstruct analog signals.
- To analyze digital signal processing systems using DTFT and Z transform.
- To design FIR and IIR filters to meet specific magnitude and phase requirements
- To learn how to use a powerful general-purpose mathematical tool such as MATLAB to design and simulate DSP systems.
- To understand the architecture of a digital signal processor and some programming issues in fixed-point digital signal processor in real-time implementation.
- To design real-time signal processing algorithms using a DSP processor.

Course Outcomes

- Explain the basic concepts of signal processing.
- Characterise signals and systems in discrete time, including use of the z-transform.
- Use Fourier transform and convolution to filter signals.
- Design FIR and IIR filters to satisfy a desired frequency response.
- Explain the role of the window function and describe its influence on FIR filters.
- Design IIR filters on the basis of an analogue design.
- Use MATLAB for DSP system analysis and design.
- Design and evaluate a real-time DSP systems.

Syllabus

Introduction, signals, systems, examples, classification of signals, basic operations on signals; folding, shifting, time scaling, classification of systems, linear and nonlinear, time variant and invariant, causal and noncausal, stable and unstable systems.

DSP Advantages and disadvantages; Quantisation and sampling; Aliasing; Anti-aliasing and reconstruction filters.

Time domain analysis of discrete signals and systems: Impulse response, Linear Time Invariant (LTI) systems, Convolution sum.

Frequency domain analysis of systems using Fourier transform .

Difference equations and z transforms.

Finite and infinite impulse response filters, FIR and IIR,

Poles, zeros and frequency response.

Design of Digital Filters-Design of nonrecursive and recursive digital filters.

Hardware & software implementations.

Some Typical DSP Applications Speech recognition; Control; Image recognition; Radar; Room response analysis, signal compression.

DSP Hardware: A/Ds, D/As and over-sampling.

Microprocessors; DSP processors; Fixed and floating point comparisons.

Programming and debugging techniques.

Lab Practice

- Verification of sampling theorem.
- Linear and circular convolution of sequences.
- Auto and cross correlation of sequences.
- Solving a difference equation.
- Computation of N point DFT of a sequence, plot magnitude and Phase spectrum
- DFT computation of square pulse and Sinc function etc.
- Design and implementation of FIR filter to meet given specifications (using different window techniques).
- Design and implementation of IIR filter to meet given specifications.
- Introduction to Code Composer Studio-I.
- Introduction to the Addressing Modes.
- Audio Codec and its Applications.
- Real Time Data Exchange.

Textbooks/References

1. Proakis John G and Manolakis Dimitris G, Digital Signal Processing: Principles, Algorithms and Applications, 4th Edition, Prentice-Hall, 2006
2. Oppenheim Alan V, Schafer Ronald W and Buck John R, Discrete Time Signal Processing, 3rd Edition, Pearson Education, 2009.
3. Prandoni Paolo and Vetterli Martin, Signal Processing for Communication, 1st Edition, EPFL Press.
4. Mitra Sanjit K, Digital Signal Processing : A Computer Based Approach, 4th Edition, McGraw Hill, 2011.
5. Kuo Sen M, Lee Bob H and Tian Wenshun, Real-Time Digital Signal Processing: Implementations and Applications, 2nd Edition, John Wiley, 2006.

6. Lapsley Phil, DSP Processor Fundamentals: Architectures and Features, IEEE Press, 1997

7. Ackenhusen John G, Real Time Signal Processing: Design and Implementation of Signal Processing Systems, Prentice-Hall, 1999.

IEG313 Quantum Computing for Engineers: Theory and Applications [3-0-2-4]

Course Objectives

- Introduce the fundamental principles of quantum mechanics relevant to computing.
- Provide an understanding of quantum computing models and algorithms
- Develop knowledge of quantum gates, circuits, and error correction techniques.
- Familiarize students with quantum programming using modern tools.
- Explore quantum software engineering concepts for practical implementation

Course Outcomes

After successful completion of the course, students should be able to:

- Introduce the fundamental principles of quantum mechanics relevant to computing.
- Provide an understanding of quantum computing models and algorithms.
- Develop knowledge of quantum gates, circuits, and error correction techniques
- Familiarize students with quantum programming using modern tools.
- Explore quantum software engineering concepts for practical implementation

Syllabus

Fundamentals of Quantum Mechanics: Introduction to Quantum Mechanics, Postulates of Quantum Mechanics, Time evolution - Schrodinger equation - Particle in a box, infinite well - Tunneling, Wave nature of Particles, Quantum States and Qubits, Superposition and Entanglement, Measurement in Quantum Systems

Introduction to Quantum Computing: Classical vs. Quantum Computing, Mathematical representations for Quantum Computing, Quantum Bits (Qubits) and Quantum Registers, Quantum Parallelism and Quantum Speedup, No-Cloning Theorem, Quantum Decoherence and Noise, Platforms for Exploring Quantum Computing (Classiq, IBM Quantum, Google Quantum)

Quantum Gates and Circuits: Quantum Gates: Pauli, Hadamard, CNOT, Phase, and Toffoli Gates, Quantum Circuits and Their Representations, Reversible and Universal Gate Sets, Quantum Error Correction Techniques, Quantum Teleportation and Superdense Coding

Quantum Algorithms and Programming: Introduction to Quantum Algorithms, Deutsch Josza algorithm, Shor's Algorithm for Factorization, Grover's Algorithm, Variational Quantum Algorithms, Introduction to Quantum Programming Languages (Qiskit, Cirq, PennyLane, etc.)

Quantum Software Engineering and Applications: Quantum Software Development Life Cycle, Quantum Computing Frameworks and Simulators, Quantum Error Correction Codes, Quantum Machine Learning: Introduction and Use cases, Hardware Implementations and Current Limitations, Future Trends in Quantum Computing. **Lab Practice**

- Hands-on Implementation of Quantum Gates and Circuits
- Bloch Sphere Visualization
- Simulating Quantum States and Superposition
- Demo with Quantum Odyssey
- Implementing Grover's and Shor's Algorithms
- Writing Quantum Programs in Qiskit, Cirq, PennyLane.
- Familiarization of Azure SDK, Ocean SDK, etc.
- Debugging and Optimizing Quantum Circuits
- Working with Real Quantum Hardware (Cloud-based IBM Quantum, Google Quantum)
- Applications: Portfolio Optimization, Quantum Teleportation and other trending applications.

Textbooks/References:

1. Yanofsky, Noson S., and Mirco A. Mannucci. Quantum computing for computer scientists. Cambridge University Press, 2008.
2. Hiu Yung Wong, Introduction to quantum computing: from a layperson to a programmer in 30 steps. Springer Nature, 2023.
3. Thomas G. Wong, Introduction to classical and quantum computing. Omaha, NE, USA: Rooted Grove, 2022.
4. Robert S. Sutor, Dancing with Qubits: How quantum computing works and how it can change the world. Packt Publishing Ltd, 2019.
5. Nielsen, Michael A., and Isaac L. Chuang. Quantum computation and quantum information. Cambridge university press, 2010.
6. Robert Loredo, Learn Quantum Computing with Python and IBM Quantum Experience: A hands-on introduction to quantum computing and writing your own quantum programs with Python. Packt Publishing Ltd, 2020.
7. Willi-Hans Steeb, and Yorick Hardy. Problems and solutions in quantum computing and quantum information. World Scientific Publishing Company, 2018.
8. Anshul Saxena, Javier Mancilla, Iraitz Montalban, and Christophe Pere. Financial Modeling Using Quantum Computing: Design and manage quantum machine learning solutions for financial analysis and decision making. Packt Publishing Ltd, 2023.

IMA314 Optimization Techniques for Data Science [3-0-0-3]

Course Prerequisite

- A preliminary knowledge of linear algebra is required for concept formulation.
- Knowledge of multivariate calculus is needed.
- Knowledge of basic concept of probability is required.
- Knowledge of programming language is essential for implementation.

Course Objectives

- To provide a conceptual understanding of optimization algorithms.
- To provide an ability to model a data science problem into an optimization problem.
- To provide an ability to select a specific optimization algorithm for solving a specific optimization problem.
- To provide new insights for observing machine learning and deep learning concepts through optimization point of view.

Course Outcomes

- Students will have a deep understanding of optimization algorithms.
- Students will be able to model a data-science problem as an optimization problem.
- Students will be able to use a specific optimization algorithm for a specific data science problem.
- Students will be familiar with the application of different optimization techniques for nonlinear regression, logistic regression, deep neural network, and support vector machine models.

Syllabus: Mathematical background: Vectors, matrices, multivariate calculus, derivatives, gradient, hessian, Taylor series, local and global minima or maxima, necessary and sufficient conditions for local and global minima.

First order methods for optimization: Gradient decent, stochastic gradient decent, batch gradient decent, convergence of gradient decent, conjugate gradient, Nesterov accelerated gradient decent, momentum, Nesterov momentum, Adam algorithm, RMSprop, hyper-gradient decent, Adadelta, Adagrad, line search methods, Wolfe conditions, Goldstein conditions, sufficient decrease and backtracking.

Second order methods for optimization: Newton's algorithm, quasi-newton's algorithm, BFGS method, properties of BFGS methods, global convergence of BFGS method, super-linear convergence of BFGS methods, inexact newton methods for large scale optimization, Limited-memory quasi-newton methods, limited-memory BFGS.

Constrained Optimization: Equality and inequality constraints,

Lagrange multipliers, duality, penalty methods, augmented Lagrange methods. Convex optimization: Convex set, convex function, unconstrained and constrained convex optimization problems, linear and quadratic optimization problems, duality.

Textbooks/References

1. Bertsekas, D. P. (1999). Nonlinear programming/Dimitri P. Bertsekas. Athena Scientific, Belmont, Mass., 2nd ed.. edition.
2. Bertsekas, D. (2015). Convex optimization algorithms. Athena Scientific.
3. Bertsekas, D. P. (2014). Constrained optimization and Lagrange multiplier methods. Academic press.
4. Boyd, S., Boyd, S. P., & Vandenberghe, L. (2004). Convex optimization. Cambridge university press.
5. Nocedal, J., & Wright, S. J. (Eds.). (1999). Numerical optimization. New York, NY: Springer New York.
6. Kochenderfer, M. J., & Wheeler, T. A. (2019). Algorithms for optimization. Mit Press.

IHS311 Human Resource Management [1-0-0-1]

Course Objectives

- To provide participants a synthesized framework of Human Resources theory & Practice.
- To impart fundamentals of HR Practices in Organizations.
- Learn to align HR systems with the strategic business objectives of a firm.

Course Outcomes

- Students will become familiar with the fundamentals of HR.
- It will equip them the tools and techniques of modern Human Resources Management.

Syllabus:

Introduction to Human Resource Management-Importance-Scope and Objectives. Evolution. Line and Staff aspects of HRM, Line managers; Duties of Human Resources Managers-Human capital management.

Job analysis: Methods for collecting Job Analysis Information, Human Resource Planning and Recruiting: The Recruitment and Selection process- Planning and Forecasting, Internal and External sources of candidates, Writing Job Description & Job Specification, Managing HR in challenging times- Testing and Selection: Basic testing concepts, Types of Tests. Interview: Process and Types, Guidelines for Interviews.

Training & Performance Management: Orientation, Training Process, Training Needs Analysis, Training Techniques-On -the-Job & Off-the -Job Training Methods, OJT Process, Training Evaluation. Management Development Programs: Case Study and other Modern Training Methods. Performance

Management & Appraisal: Process and Techniques. Career Planning and Management Concepts.

Compensation Establishing Pay Rates: Steps, Job Evaluation -Wage and Salary administration- Steps and factors affecting, Incentives Benefits and services: Statutory Benefits - Non-statutory Benefits - Insurance Benefits -Retirement Benefits, Flexible Benefits Programs. ESOPs, QWL.

Industrial Relations & Trends in HR Industrial relations: Significance, Objectives, Approaches. Industrial Disputes-Causes, Forms, Preventive Machinery. Collective Bargaining: Basic Concepts. Long term settlements: Cases in India. Trade unions: Definition, Objectives, Functions Social Security in India, Employee welfare, Grievance Handling and Discipline-Sources and forms of Grievances -Grievance Procedure, Disciplinary Procedure. Participative Decision-making process - Role of quality circle in TQM. Strategic Human Resources Management, Strategic HRM tools. An over view of HR Analytics

Textbooks/References

1. Brealey, Richard A and Stewart CMyers. Principles of Corporate Finance. McGraw Hill India, 2012.
2. Chandra Prasanna, Financial Management- Theory & Practice, Tata McGraw Hill, 2014..
3. James C Vanhorne, John M WachowiczJr, Fundamentals of Financial Management, Pearson Education Limited, New Delhi.
4. Lawrence J Gitman, Principles of Managerial Finance, Pearson Education limited. New Delhi.
5. Pandey IM, Financial Management. Vikas Publishing House, 2009.
6. Reddy, G Sudarsana, Financial Management, Himalaya Publishing House, 2011.
7. Van Horne James, Financial Management Policy, Prentice Hall India

IHS312 Financial Management and Accounting [1-0-0-1]

Course Objectives

- To familiarize participants with fundamentals of Financial Management in an Organization.
- To provide the participants various techniques in Financial Management.
- To give an overview of the emerging financial issues facing an Organization.

Course Outcomes

- Students will become familiar with accounting standards and financial management.
- It will equip the skills for better financial management.

Syllabus

Introduction to Finance: Introduction to Financial management: Business Finance- Concept, types and scope. Financial management: objectives, functions and scope - Interface of financial management with other functional areas. Role of finance manager- Financial forecasting - Financial planning. Risk and Return concept: - Relationship between risk and return - Risk Diversification.

Time Value of Money & Investment Decisions: Process of compounding - Process of discounting - Future value of Single cash flow and annuity - Present value of a single cash flow and annuity. Investment Decisions: - Capital budgeting - Process of capital budgeting - selection of projects - Estimation of cash flows - Payback and Discounted payback period - Accounting rate of return- NPV - IRR - Capital Budgeting decisions under risk - Capital Rationing - Project selection under rationing.

Finance Decisions: Financing and Capital Structure Decision: Sources of Finance: External and Internal financing. Cost of different sources of capital - Weighted average cost of capital (WACC) and Marginal cost of capital. Capital structure decisions - meaning and pattern- Theories of capital structure- Net income approach - Net operating income approach- Traditional approach-MM approach - Optimum capital structure. Leverage - operating, financial and composite leverage.

Dividend Decisions: Dividend policy - dividend and its forms - objectives of dividend policy- relevance and irrelevance. Theories of dividend decisions: Walter's Approach - Gordon's Approach - MM Approach.

Working Capital: Management of Working capital: Meaning and Need of Working capital - factors affecting composition of working capital - Inter dependence among components of working capital - Estimation of working capital - Cash management- Cash flow statement and fund flow statement- Receivables management.

Textbooks/References:

1. Brealey, Richard A and Stewart CMyers. Principles of Corporate Finance. McGraw Hill India, 2012.
2. Chandra Prasanna, Financial Management- Theory & Practice, Tata McGraw Hill, 2014..
3. James C Vanhorne, John M WachowiczJr, Fundamentals of Financial Management, Pearson Education Limited, New Delhi.
4. Lawrence J Gitman, Principles of Managerial Finance, Pearson Education limited. New Delhi.
5. Pandey IM, Financial Management. Vikas Publishing House, 2009.
6. Reddy, G Sudarsana, Financial Management, Himalaya Publishing House, 2011.
7. Van Horne James, Financial Management Policy, Prentice Hall India

IHS313 Operations and Supply Chain Management [1-0-0-1]

Course Objectives

- To gain a working understanding of logistics principles and to expose students to the basic concepts the language of logistics and supply chain management.
- To refine applied data analysis skills by analyzing and using supply chain data to evaluate supply chain performance and to make business decisions.
- To understand current challenges faced by supply chain professionals and to provide a basis for thinking through these challenges.

Course Outcomes

- Students will become familiar with basic concepts of Supplychain Management.
- Students will get a better understanding on web based and electronic applications of supplychain management.

Syllabus:

Introduction to Production & Operations Function:- Scope of Production and Operations Management - Interaction of Operations Management with other functional areas of Management - Manufacturing and Non Manufacturing operations and their Classifications - Operations Strategy: Elements of Operations Strategy - 5Ps of Operations.

Production & Operations Planning, Manufacturing System, Layout: Production & Operations Planning and Control, Role of Production Planning & Control in Operations Management - Plant Location & Layout: Steps in location selection - Factors influencing Layout - Principles of Layout - Layouts by Products and Process - Hybrid Layout -Design of Operations Systems: Aggregate planning and Master Scheduling, MRP, CRP. Line Balancing & Sequencing - Capacity Planning.

Materials Management & Vendor Management:Material Handling: Material Handling Principles - Types - Selection & Design of Material Handling System. Materials Management - Functions - Material planning and Budgeting - Value Analysis - Purchase functions and Procedure - Inventory control - Types of Inventory - Safety stock - Inventory Control Systems - Economic Order Quantity (EOQ) - Perpetual - Periodic - Just In Time (JIT) - Managing Vendors; Vendor Analysis, Rating and Selection - Procedure and Criteria.

Work Study & Maintenance - Work study, Time and Method study: Definition - Importance - Aims and Procedures - Implications on Productivity - Work measurement - Work sampling - Work environment - Industrial safety - Value analysis. Basics of Maintenance Management - Maintenance Decisions.

Supply Chain Management & Lean Systems: Supply Chain Management -Concept of Supply chain, Stages and flows in Supply chain, Terminology in Supply chain management -Supply chain disruption- Bull Whip effect. Lean Systems - Basic understanding about Lean concepts- Pull and Push

systems, Jidoka, Poke-Yoke, 5S, Total Preventive Maintenance (TPM), Toyota Production System, Kanban System.

Textbooks/Reference:

1. Lee J. Krajewski et al, Operations Management, Process and Supply chains. 11th Edition Pearson India Education Services Ltd. India
2. Russel & Taylor, Wiley, Management, Quality and Competitiveness in a Global Environment, Fifth Edition, India Edition
3. BuffaSarin, Wiley, Modern Production and Operations Management, India Edition
4. KanishkaBedi, Production and Operations Management, Oxford University Press.
5. Aswathappa K and ShridharaBhat K, Production and Operations Management, Himalaya Publishing House, Revised Second Edition, 2008.
6. Pannierselvam R, Production and Operations Management, Prentice Hall India, Second Edition, 2008.
7. Mahadevan B, Operations Management Theory and Practice, Pearson Education, 2007.

SEMESTER VI

CSE321 Embedded Systems Design [3-0-2-4]

Course Objectives

To impart knowledge on

- The embedded systems and their design
- Programming embedded micro-controllers
- Real-time applications using embedded systems.
- Tasks scheduling in a multiprocessor environment

Course Outcomes

After successful completion of the course, students should be able to:

- Understand the features of micro-controllers, ARM processors and their applications.
- Classify real-time systems like hard, firm and soft RTOS and know their applications.
- Program latest ARM processors
- Create an embedded system for an use-case

Syllabus:

Classification of microprocessors, microcontrollers, and ARM processors. General purpose computers vs. embedded systems. CISC and RISC concepts, Big-endian and Little-endian processors, parallelism via pipelining, features of Von Neumann and Harvard architecture. A walkthrough on micro-controller: 8051, architecture, features: parallel I/O interface, serial I/O interface, Counter/Timer, interrupt control mechanism, programming examples and limitations. Introduction to ARM7TDMI-S processor [LPC2148], architecture and features: Registers, Multiplier, Barrel shifter, GPIO, Serial communication interfaces, UART and timers.

Instruction set: Data transfer instructions, arithmetic, logic operations, shift and rotate instructions, bit-field processing instructions, compare and test, program flow control, exception-related instructions. Access to special registers. Use of assembly codes in projects, interaction between C and assembly, structure of an assembly function, simple example with ARM tool chains, example with GNU tools for ARM embedded processors, accessing special registers, displaying a string. Simulation example using Proteus Design Suite.

Process, multitasking, process management: periodic and aperiodic task systems EDF and RMS algorithms, memory management, I/O and file systems management, POSIX. Embedded Computing Systems on FPGAs: SRAMs based FPGAs, Flash based FPGAs, OS support, Challenges in Embedded Computing Design on FPGAs.

Lab Practice

Microcontroller based:

- LED blinking using 8051 microcontroller
- Display characters/string on LCD using 8051 microcontroller.

ARM processor based:

- LED blinking
- Interfacing of LCD
- Interfacing of 7-segment display and displaying numbers
- Interfacing of keyboard matrix
- Interfacing of stepper motor
- Establish I2C communication between the ARM processor and an external device such as an EEPROM

Textbooks/Reference:

1. C. Hamacher, Z. Vranesic, S. Zaky, and N. Manjikian, Computer Organization and Embedded Systems, 6th ed. New York, NY, USA: McGraw-Hill, 2012
2. Yiu, J. The Definitive Guide to ARM Cortex-M3 and Cortex-M4 Processors. 3rd ed.; Newnes: Cambridge, MA, 2013
3. Wolf, M. Computers as Components: Principles of Embedded Computing System Design. 5th ed.; Elsevier: Cambridge, MA, 2022
4. P. Marwedel, Embedded System Design: Embedded Systems Foundations of Cyber-Physical Systems, and the Internet of Things, 4th ed. Cham, Switzerland: Springer, 2021
5. T. Noergaard, Embedded Systems Architecture: A Comprehensive Guide for Engineers and Programmers, 2nd ed., Elsevier Embedded Technology Series, Newnes Publication, 2012
6. K. Iniewski, Embedded Systems: Hardware, Design, and Implementation, Wiley & Sons, Inc., 2013
7. R. Kamal, Embedded Systems: Architecture, Programming and Design, 3rd ed., McGraw Hill Education (India), 2014
8. M. Barr, Programming Embedded Systems in C and C++. Sebastopol, CA, USA: O'Reilly Media, 1999
9. S. R. Ball, Embedded Microprocessor Systems: Real World Design, 2nd ed., Newton, Mass., USA, 2002

CSE322 Cloud Computing [3-0-2-4]

Course Prerequisites

The participants should have prior knowledge on the following topics/courses:

- Computer Networks
- Parallel and Distributed Computing
- Fundamentals of Programming

Course Objectives

- To gain knowledge on virtualization techniques.
- To frame VM clusters.
- To learn OS-level virtualization tools, including dockers.

- To understand the working methodology of existing clouds, such as, Amazon, Opennebula, GCE, and so forth.
- To learn how to program clouds using new programming models.

Course Outcomes

The main outcomes of this course are listed as follows:

- The students will learn the base technologies for cloud.
- Apply appropriate cloud services for their applications.
- Design cloud services using goLang or nodejs.
- Learn how to program public clouds such as AWS or GCE.

Syllabus: Base Technologies-Review: Introduction, Grid Computing, Cluster, P2P computing, and so forth, System Models for Distributed and Cloud computing.

VM server consolidation, VirtualBox, VMWare Vsphere - Datacenter Automation.

Virtualization: Virtualization concepts, levels of Virtualization, VM Tools and mechanisms, Virtualization of CPU, Memory, and I/O devices, VM server consolidation, VirtualBox, VMWare Vsphere - Datacenter Automation, Docker containers, orchestration platforms – Kubernetes, Docker Swarm.

Cloud Infrastructure / Architectures: Design Challenges of Clouds, Public cloud platforms – GCE, AWS, Azure, Resource Management in Clouds, Private cloud environments - openstack and opennebula, security aspects of clouds, Storage aspects of clouds, introduction to programming models.

Advanced topics: Cloud Automation Tools – CI/CD pipelines, Terraform, Ansible, CloudFormation, Google Cloud Deployment manager, Jenkins, Puppet, Chef, Spinnaker, GitOps tools-ArgoCD, GitHub Actions, Devops, Cloud Networking - SDN, HPC in the cloud, IoT cloud, Serverless Cloud platforms.

Lab Practice

- VMs using virtualbox, Golang programming, Vshpere, Live migration of VMs using Vshpere, Docker, Kubernetes, AWS EC2 or GCE or Azure, AWS S3, AWS Lambda or Google Functions or Azure functions, Cloud Automation tools – Gitops, DevOps.

Textbooks/References

1. Kai Hwang, Geoffrey C. Fox, Jack K. Dongarra, Distributed and Cloud Computing: From parallel processing to Internet of Things, Morgan Kauffmann 2013.
2. William Stallings, Foundations of Modern Networking: SDN, NFV, QoE, IoT, and Cloud, Pearson publishers, 2016.
3. Jonathan Baier, Getting Started with Kubernetes: 2nd Edition, Packt publishers, 2015.
4. Vaishnavi P. Cloud Computing: An Efficient Load Balancing and Resource Provisioning in the Cloud Environment, Scholars press, 2023.

5. Rajkumar Buyya, James Broberg, Andrzej Goscinski, Cloud Computing: Principles and Paradigms, 2011 John Wiley & Sons, Inc
6. Kevin Hoffman and Dan Nemeth, Cloud Native Go: Building Web Applications and Microservices for the Cloud with Go and React (Developer's Library), Pearson publishers, 2016.
7. Bob Familiar, Microservices, IoT, and Azure: Leveraging DevOps and Microservice Architecture to deliver SaaS Solutions, Apress publishers, 2015.
8. Dirk Slama, Frank Puhlmann, Jim Morrish, and Rishi M. Bhatnagar, Enterprise IoT: Strategies and Best practices for Connected products and services, O'Reilly publishers, 2015.
9. Oluyemi James Odeyinka, AWS Cloud Automation: In-depth guide to automation using Terraform infrastructure as code solutions, Bpb publishers, 2024.

CSS321 Data Warehousing and Data Mining [3-0-2-4]

Course Objectives

- To introduce the basic concepts of Data Warehouse and Data Mining techniques.
- Examine the types of the data to be mined and apply pre-processing methods on raw data.
- Apply basic classification, clustering and outlier analysis on a set of data.

Course Outcomes

- Create a data warehouse and process raw data to make it suitable for various data mining algorithms.
- Discover and measure interesting patterns from different kinds of databases.
- Apply the techniques of clustering, classification, association finding, feature selection and visualization to real world data.

Syllabus

Overview of Knowledge extraction, Data Warehousing concepts and Architecture, Online Analytical Processing (OLAP) – OLAP and Multidimensional Data Representation, Data cube technologies, Business Intelligence.

Data Mining: - Data Mining Functionalities – Data Pre-processing – Data Cleaning – Data Integration and Transformation – Data Reduction – Data Discretization and Concept Hierarchy Generation. Association Rule Mining.

Classification and Prediction:-Issues Regarding Classification and Prediction – Classification by Decision Tree. Introduction – Bayesian Classification – Rule Based Classification – Classification by Back propagation – Support Vector Machines – Associative Classification – Lazy Learners – Other Classification Methods.

Cluster Analysis:- Types of Data in Cluster Analysis, Model-Based Clustering Methods, Hierarchical and Partitioning methods. Outlier Analysis. Applications and trends in Data Mining: Mining Text and Web data.

Lab Practice

- Basics of R
- R for Data Science
- Data Preprocessing and cleaning
- Data Reduction
- Association rule mining
- Classification and Prediction
- Cluster Anlayis, Outlier analysis
- Mining web

Textbooks/References:

1. Alex Berson, Stephen J. Smith, "Data Warehousing, Data Mining, & OLAP", Tata McGraw- Hill, 2004.
2. Jiawei Han. Data Mining: Concepts and Techniques. Morgan Kaufmann Publishers
3. Anahory and Murray .,Data warehousing in the real world ,Pearson Education/Addison Wesley.
4. Berry Micheal and Gordon Linoff,. Mastering Data Mining. John Wiley & Sons Inc.
5. Margaret H. Dunham Data Mining: Introductory and Advanced Topics. Prentice Hall.

CSS322 Machine Learning [3-0-0-3]

Course Prerequisites

- Calculus & Linear Algebra
- Programming & Data Structure Course

Course Objectives

- To provide an in-depth introduction to supervised, unsupervised and reinforcement learning algorithms.
- To provide a sound mathematical understanding of ML algorithms
- To design and implement machine learning solutions to classification, regression, and clustering problems.

Course Outcomes

- Develop an appreciation for what is involved in learning from data.
- Understand a wide variety of learning algorithms and their associated mathematical foundations to appreciate these algorithms.

- Understand how to apply a variety of learning algorithms to data.
- Understand how to perform evaluation of learning algorithms and model selection.

Syllabus

Review of linear algebra, optimization and probability: Matrices, Eigen values and vectors, gradient, hessian, least squares, optimization; random variables and distributions.

Definitions, goals and history of Machine Learning; Introduction, linear classification; Classification errors; Regression Techniques, Model Selection (Empirical risk minimization (ERM), Structural risk minimization (SRM), cross-validation, Regularization)

Supervised learning: generative/discriminative learning, parametric/non-parametric learning, neural networks, Gradient Descent, support vector machines, large margins, Boosting.

Unsupervised learning: clustering, dimensionality reduction, kernel methods; learning theory, bias/variance trade-offs; VC theory; large margins.

Reinforcement learning and adaptive control. Applications of machine learning.

Textbooks/References

1. Mitchell, Tom. Machine Learning. New York, NY: McGraw-Hill, 1997.
2. Bishop, C. ,M., Pattern Recognition and Machine Learning, Springer, 2006
3. P. Langley, Elements of Machine Learning, Morgan Kaufmann, 1995.
4. Hastie, T., R. Tibshirani, and J. H. Friedman. The Elements of Statistical Learning: Data Mining, Inference and Prediction, Second Edition, Springer, 2009
5. MacKay, David. Information Theory, Inference, and Learning Algorithms. Cambridge, UK: Cambridge University Press, 2003.
6. Foundations of Machine Learning, by Mehryar Mohri, Afshin Rostamizadeh, and Ameet Talwalkar

ISCXXX Introduction to Network Science Analytics [3-0-0-3]

Course Objectives

- Understand the components of the network.
- Model and visualize the network.
- Understand the role of semantic web in the social network.
- Familiarize the security concepts of the social network.
- Apply and analyze the various applications.

Course Outcomes

- Illustrate the basic concepts of network science.
- Design and apply network models.
- Develop social network applications using visualization tools.
- Use the security features in social and information network.
- Apply semantic web techniques in social network.

Syllabus

Introduction: Introduction to network analysis – Fundamental concepts in network analysis – social network data – notations for social network data – Graphs and Matrices.

Measures & Metrics: Strategic network formation - network centrality measures: degree, betweenness, closeness, eigenvector - network centralization-density -reciprocity - transitivity - ego network - measures for ego network - dyadic network - triadic network - cliques - groups- clustering -search.

Community networks: Community structure - modularity, overlapping communities -detecting communities in social networks – discovering communities: methodology, applications - community measurement - evaluating communities – Applications.

Models: Small world network - Watts-Strogatz networks - Statistical Models for Social Networks – Network evolution models: dynamical models, growing models - Nodal attribute model: exponential random graph models – Preferential attachment -Power Law - random network model: Erdos-Renyi and Barabasi Albert-Epidemics - Hybrid models of Network Formation.

Visualization of social networks – novel visualizations and interactions for social networks.

Semantic Web: Modelling and aggregating social network data – developing social semantic application – evaluation of web-based social network extraction – Data Mining – Text Mining in social network – Tools -case study.

Security & Applications: Managing Trust in online social network – Security and Privacy in online social network – security requirement for social network in Web 2.0.

Textbooks/References

1. Tanmoy Chakraborty Social Network Analysis, Wiley India Pvt Ltd, 2021.
2. Nilanjan Dey, Samarjeet Borah, Rosalina Babo, Amira S. Ashour, Social Network Analytics.
3. Computational Research Methods and Techniques, 2018.
4. Stanley Wasserman, Katherine Faust, Social network analysis: Methods and applications, Cambridge university press, 2012.
5. John Scott, Social network analysis, 3rd edition, SAGE, 2013.
6. Charu Aggarwal, Social Network data analysis, Springer, 2011.

IOEXXX Introduction to DevOps and Microservices [3-0-0-3]

Course Objectives

- To understand and explain the concepts of business needs for DevOps and Microservices.
- To gain knowledge in DevOps design issues and adopting DevOps in business use cases.
- To develop and integrate Microservices using DevOps.
- To deploy the DevOps automation for different use cases.

Course Outcomes

- Understand the key concepts and principles of DevOps and Microservices.
- Familiar with DevOps automation tools.
- Apply Microservices in DevOps.
- Apply DevOps for Business Analytics.

Syllabus

Introduction to DevOps: DevOps: An overview - Understanding the Business Needs for DevOps - History of DevOps - DevOps and Software Development Life Cycle – Waterfall Model - Agile Model – DevOps Lifecycle - Process and Technology in DevOps - DevOps Myths – Barriers.

DevOps Adoption and Tools - Plan and Measure, Develop and Test (collaborative and continuous), Release and Deploy Monitor and Optimize. Tools: distributed version of control tool Git- Automation testing tools-Selenium – report generation –TestNG – User Acceptance Testing – Jenkins.

Microservices: Introduction to Microservices, Architecture: Monolith and Microservices- Benefits,Challenges,Characteristics: Coordination model - Building and testing - Deployment pipeline - Development and Pre-commit Testing - Build and Integration Testing - Continuous integration - Monitoring.

Integration of Microservices and DevOps – Build a DevOps platform - Benefits of combining DevOps and Microservices-working of DevOps and Microservices in Cloud environment (Case studies: Openstack, GCP and AWS).

Devops Automation: Infrastructure Automation- Configuration Management - Deployment Automation - Performance Management - Log Management -Monitoring.

Textbooks/References

1. Joyner Joseph, “Devops for Beginners”, 1st Edition, Mihails Konoplovs publisher, 2015.
2. Stephen Flemming, “DevOps And Microservices Handbook: Non-Programmer’s Guide to DevOps and Microservices”, 2018

3. Michael Hüttermann, DevOps for Developers, 1st Edition, APress, e-book, 2012.
4. James A Scott, A Practical Guide to Microservices and Containers, MapR Data Technologies
5. Ebert, C., Gallardo, G., Hernantes, J., & Serrano, N. (2016). DevOps. Ieee, 33(3), 94-100.

SEMESTER VII

CSE411 Computer Vision: Principles and Processes [3-0-2-4]

Course Prerequisites

- To explore fundamental topics in the field ranging from low-level feature extraction to high-level visual recognition.
- To understand and apply fundamental mathematical and computational techniques in computer vision.

Course Objectives

- To understand the basic signals in the field of biomedical.
- To study origins and characteristics of some of the most commonly used biomedical signals, including ECG, EEG, evoked potentials, and EMG.
- To understand Sources and characteristics of noise and artifacts in bio signals.
- To understand use of bio signals in diagnosis, patient monitoring and physiological investigation
- To explore research domain in biomedical signal processing

Course Outcomes

- Develop methods that enable a machine to “understand” or analyze images and videos.
- Apply a variety of computer techniques for the design of efficient algorithms for real-world applications.

Syllabus

Introduction: Human visual recognition system; Why is vision difficult? Retina, LGN and primary visual cortex; Computational models of human visual object recognition; Representation, Learning, Detection, Recognition.

Features and filters: low-level vision: Linear filters, Edges and contours, Binary image analysis, Background subtraction, Texture, Motion and optical flow.

Grouping and fitting: mid-level vision: Segmentation and clustering algorithms, Hough transform, Fitting lines and curves, Robust fitting, RANSAC, Deformable contours, Interactive segmentation.

Multiple views: Local invariant feature detection and description, Image transformations and alignment, Planar homography, Epipolar geometry and stereo, Object instance recognition.

Recognition: high-level vision: Object/scene/activity categorization, Object detection, Supervised classification algorithms, Probabilistic models for sequence data, Visual attributes, Active learning, Dimensionality reduction, Non-parametric methods and big data, Deep learning, convolutional neural networks.

Applications: Optical character recognition, face detection and recognition, motion estimation, human tracking, and

gesture recognition.

Lab Practice

- Projective Geometry, Transformation of 2D and 3D, Internal Parameters. Local Feature Detectors and Descriptors: Hessian corner detector, Harris. Corner Detector, LOG detector, DOG detector, SIFT, PCA-SIFT, GLOH, SURF, HOG, Pyramidal HOG, PHOW-Calibration Methods: Linear, Direct, Indirect and Multiplane methods. Edge Detection – Roberts and Sobel, Histogram Thresholding Segmentation, Morphological Filters

Textbooks/References

1. Richard Szeliski. "Computer vision: algorithms and applications". Springer, 2021.
2. Valliappa Lakshmanan, Martin Görner, Ryan Gillard, Practical Machine Learning for Computer Vision, 2021.
3. Laganier, Robert. "OpenCV computer vision application programming cookbook second edition". Packt Publishing Ltd, 2014.
4. David A. Forsyth and Jean Ponce, "Computer Vision: A Modern Approach", first edition, Prentice Hall, 2015.
5. Md Atiqur Rahman Ahad. "Computer vision and action recognition: a guide for image processing and computer vision community for action understanding". Vol. 5. Springer, 2014.
6. DiCarlo, James J., Davide Zoccolan, and Nicole C. Rust. "How does the brain solve visual object recognition?" *Neuron* 73.3 (2012): 415-434.
7. Kristen Grauman. "Visual Recognition". Course at University of Texas Austin. Fall 2012.
8. Raquel Urtasun. "Visual Recognition". Course at Toyota Technological Institute. Winter 2012
9. Gabriel Kreiman. "Visual Object Recognition: Computational and Biological Mechanisms". Course at Harvard University. Fall 2013.
10. Rob Fergus. "Visual Object Recognition and Retrieval". Tutorial in ICML 2008.

CSE413 Speech and Natural Language Processing [3-0-2-4]

Course Objectives

- Introduction to the fundamental concepts and techniques of Natural Language Processing (NLP)
- Gain in-depth understanding of the computational properties of natural languages and commonly used algorithms for processing linguistic information
- Hands-on and Practical Applications of NLP

Course Outcomes

- Understanding of key concepts, models and challenges in Natural Language Processing

- Understand the structure of language, current techniques, strategies, and toolkits for NLP
- Be able to describe current approaches, datasets and systems for various Natural Language Processing tasks

Syllabus

NLP Concepts: What is NLP? Morphology fundamentals; Isolating, Inflectional, Agglutinative morphology; Infix, Prefix and Postfix Morphemes, Morphological Diversity of Indian Languages; Morphology Paradigms; Rule Based Morphological Analysis: Finite State Machine Based Morphology; Automatic Morphology Learning.

Approaches for NLP: NLP Applications in the real world: Machine translation (Google Translate), Natural language generation, Web Search, Spam filters, Sentiment Analysis, Chatbots, question answering, document summarization, discourse.

NLP Pipeline: Text Processing- Tokenization, lemmatization, POS tagging, noun phrase chunking, name entity recognition, co-reference resolution, parsing, text normalization, exploring text data, implementing text processing using NLTK. Text Representation-Vector space models: Basic vectorization approaches: One-Hot Encoding, Bag-of-Words, Bag-of-N-Grams, Count vectorizer, TF-IDF; Distributed Representations: Word Embeddings, Document Embeddings Going Beyond Words.

Text Classification- Existing Text Classification APIs; Machine Learning Classifiers: Naive Bayes Classifier, Logistic Regression, Support Vector Machine; Neural Network Classifiers, Deep Learning Classifiers: CNNs, LSTMs, NLP Applications. Social Media, E-Commerce and Retail, Healthcare, Finance, and Law.

Advanced Topics: Self-Attention, Transformers, Language Modeling, Masked Language Modeling, BERT, Contextual Embedding.

Lab Practice

- POS Tagging: HMM, SVM, Bi-LSTM
- Text preparation: Tokenization, Stemming, Advanced linguistic structure, Sentiment lexicons.
- Chunking, Parsing: Constituency Parsing (CP) Dependency Parsing (DP) Noun Phrase Chunking, Named Entity Recognition, Co-reference resolution Sentiment Analysis, Question Answering, Text Classification.

Textbooks/References

1. Sowmya Vajjala, Bodhisattwa Majumder, Anuj Gupta, Harshit Surana. Practical Natural Language Processing, O'Reilly Media, 2020.
2. Steven Bird, Ewan Klein, Edward Loper. Natural Language Processing with Python- Analyzing Text with the Natural Language Toolkit, O'Reilly Media, 2019. <http://www.nltk.org/book/>

3. Dan Jurafsky, James H. Martin. Speech and Language Processing, 2009. (<https://web.stanford.edu/~jurafsky/slp3/>)
4. Sarkar, Dipanjan. Text Analytics with Python: A Practitioner's Guide to Natural Language Processing, 2019, Apress; 2nd ed. edition (22 May 2019)
5. Chris Manning and Hinrich Schütze, Foundations of Statistical Natural Language Processing, MIT Press. Cambridge, MA: May 1999.

CSS411 Cryptography and Network Security [3-0-2-4]

Course Objectives

- To lay a foundation on Security in Networks, attacks, defence and Classical Cryptosystems
- To analyse various Private and Public key Cryptosystems to ensure confidentiality, Integrity and Authentication.
- To analyse various protocols to ensure Email Security and Network Security.
- To apply Cryptography in various Applications.

Course Outcomes

- Understand the fundamental concepts of Cryptography, Types of Security breaches, attacks, defence, control measures, Classical Cryptosystem.
- Compare various Private and Public key Cryptosystems to ensure confidentiality, Integrity and Authentication.
- Understand various protocols in Email Security and Network Security.
- Apply Cryptography in various Applications.

Syllabus

Introduction to Security in networks, Types of Security breaches, attacks, defence, control measures, Classifying cryptosystems, classical cryptosystems, block cipher modes of operation, DES encryption and decryption, triple DES, AES encryption and decryption.

Public Key Cryptosystem - RSA cryptosystem, Diffie-Hellman Key Exchange Algorithm, Elliptic curve cryptosystem, Message Authentication and Hash Function- MD5 message digest algorithm, Secure hash algorithm, Authentication Protocols and Digital signatures.

Kerberos - X.509 Authentication Service. – Pretty Good Privacy - Electronic Mail Security – IP Security Architecture – Web Security Considerations – Secure Socket Layer and Transport Layer Security – Secure Electronic Transaction – Firewalls – Firewall Design Principles

Applications of Cryptography- Blockchain, Bitcoin and Cryptocurrency Technologies

Textbooks/References:

1. Behrouz A. Forouzan, Debdeep Mukhopadhyay, Cryptography and Network Security, 3rd Edition, Mc Graw Hill Education, 2016.
2. Stallings W., Cryptography and Network security: Principles and Practice, 7/e, Pearson Education Asia, 2017.
3. Charles P. Pleegeer, Shari Lawrence Pleegeer, "Security in Computing", Pearson Education Asia, 4th Edition, 2009.
4. Alan T. Norman, "Blockchain Technology Explained: The Ultimate Beginner's Guide About Blockchain Wallet, Mining, Bitcoin, Ethereum, Litecoin, Zcash, Monero, Ripple, Dash, IOTA and Smart Contracts" Kindle Edition, 2017.
5. Andreas Antonopoulos, "Mastering Bitcoin: Unlocking Digital Cryptocurrencies" 1st Edition, 2014.
6. Matthew Connor, "Blockchain: Ultimate Beginner's Guide to Blockchain Technology - Cryptocurrency, Smart Contracts, Distributed Ledger, Fintech, and Decentralized Applications" Kindle Edition, 2017.

IOEXXX Data Handling Analysis and Visualization [3-0-0-3]

Course Prerequisites

- Introduction to Probability.

Course Objectives

- To expose to visual representation methods and techniques that increase the understanding of complex data.
- To quickly examine large amounts of data, expose trends efficiently, exchange ideas with key players, and influence decisions.

Course Outcomes

- Understanding of good design practices for visualization, tools for visualization of data from a variety of fields.
- The ability to read and discuss research papers from the visualization literature.
- Practical experience building and evaluating visualization systems.

Syllabus

Data Definitions and Analysis Techniques: Elements, Variables, and Data categorization, Levels of Measurement, Data management and indexing, Introduction to statistical learning and R-Programming.

Descriptive Statistics: Measures of central tendency, Measures of location of dispersions, Practice and analysis with R.

Basic Analysis Techniques: Basic analysis techniques, Statistical hypothesis generation and testing, Chi-Square test, t-Test, Analysis of variance, Correlation analysis, Maximum likelihood test, Practice and analysis with R. Data analysis techniques: Regression analysis, Classification techniques, Clustering, Association rules analysis, Practice and analysis with R.

Data Visualization: Basic Plotting: Line plot - Bar plot - Pie Chart - Scatter Plot - Histogram - Stacked Bar Charts - Sub Plots - Matplotlib, Searborn, Plotly - Seaborn Styles, Applied Visualizations: Box plot - Density Plot - Area Chart - Heat map - Tree map - Graph Networks.

Interactive Visualizations and Animations: Dynamic charts - Dynamic maps - Animation types - 2D, 3D, Motion Animation - Animation Principles - Altair Package - Statistical Visualizations.

Principles of Information Visualization: Visual Perception and Cognition - Gestalt's Principles - Tufte's Principles - Applications of Principles.

Textbooks/References:

1. Probability & Statistics for Engineers & Scientists (9th Edn.), Ronald E. Walpole, Raymond H. Myers, Sharon L. Myers and Keying Ye, Prentice Hall Inc.
2. An Introduction to Statistical Learning: with Applications in R, G James, D. Witten, T Hastie, and R. Tibshirani, Springer, 2013
3. Hastie, T., Tibshirani, R., Friedman, J.H. and Friedman, J.H., "The Elements of Statistical Learning, Data Mining, Inference, and Prediction", 2017.
4. Mining Massive Data Sets, A. Rajaraman and J. Ullman, Cambridge University Press, 2012
5. The Visual Display of Quantitative Information (2nd Edition). E. Tufte. Graphics Press, 2001.
6. Visualizing Data by Ben Fry 2016.

IOEXXX Streaming Data Analytics [3-0-0-3]

Course Prerequisites

- ICS 322 Machine Learning

Course Objectives

- To understand the theoretical foundations, algorithms, methodologies.
- To provide practical knowledge for handling and analysing streaming data.
- To implement different machine learning algorithms on streaming data.

Course Outcomes

- Recognize the characteristics of data streams that make it useful to solve real-world problems.
- Identify and apply appropriate algorithms for analysing the data streams for variety of problems.

- Practising the tools and platforms required for analysing streaming data.
- Describe and apply current research trends in data-stream processing.

Syllabus

Introduction to Data Streams: Characteristics of the data streams, Challenges in mining data streams Requirements and principles for real time processing, Concept drift Incremental learning Basic Streaming Methods, Counting the Number of Occurrence of the Elements in a Stream, Counting the Number of Distinct Values in a Stream, Bounds of Random Variables, Poisson Processes, Maintaining Simple Statistics from Data Streams, Sliding Windows, Data Synopsis.

Machine Learning from Data Streams: Decision Tree (Very Fast Decision Tree Algorithm, The Base Algorithm) Clustering (Partitioning, Hierarchical), Frequent Pattern Mining (Mining Frequent Itemset from Data Streams – Landmark window).

Complex Event Processing: Introduction to Complex Event Processing, Features of CEP, Need for CEP, CEP Architectural Layers, Scaling CEP, Events, Timing and Causality, Event Patterns, Rules and Constraint, STRAW-EPL, Event Hierarchies.

Advanced concepts on Stream Analytics: Synopsis construction in data streams - sampling methods - wavelets – sketches – histograms. Join processing in data streams - indexing and querying data streams - dimensionality reduction and forecasting on streams - distributed mining of data streams.

Apache Kafka with case study: Kafka Architecture, Installation, Configuration, Performance Tuning, Kafka Client APIs and Stream APIs, Kafka Connect API and Kafka Integration with Big data ecosystems. Case Study on Twitter Streaming, Stock Market Prediction.

Textbooks/References

1. Charu C. Aggarwal, “Data Streams: Models and Algorithms”, Kluwer Academic Publishers, 2007
2. Joao Gama, “Knowledge Discovery from Data Streams”, CRC Press, 2010
3. Byron Ellis, “Real Time Analytics: Techniques to Analyze and Visualize Streaming Data”, John Wiley and Sons, 2014.
4. Shilpi Saxena, Saurabh Gupta, “Practical Real-time Data Processing and Analytics”, Pack publishing 2017
5. Neha Narkhede, Gwen Shapira, Todd Palino “Kafka: The Definitive Guide: Real-Time Data and Stream” 2017, O’Reilly

IOEXXX Dependable and Secure AI [3-0-0-3]

Course Prerequisites

- Machine Learning
- Artificial Intelligence

Course Objectives

- AI and machine learning systems.
- To make them dependable, such as explainability, interpretability, safety etc.

Course Outcomes

- Assess the dependability of AI systems.
- Develop explainable, robust, and safe AI models.

Syllabus

Introduction: Overview, Motivation, Challenges – medical and surveillance Explainable AI: Accuracy-explainability Tradeoff, Interpretability Problem, Predictability, Transparency, Traceability, Causality, Reasoning, Attention and Saliency.

Interpretable AI: Prediction Consistency, Application Level Evaluation, Human Level Evaluation, Function Level Evaluation .

Adversarial Robustness: Adversarial Attacks and Defences. Trustworthy AI: Integrity, Reproducibility, Accountability.

Bias-free AI: Accessibility, Fair, Data Agnostics Design, Disentanglement.

Privacy Preserving AI: Federated Learning, Differential Privacy and Encrypted Computation.

Verified AI: Environment and Specification Modeling, Design with Formal Inductive Synthesis, Evaluation Platforms for AI Safety.

Textbooks/References

1. J. PEARL (2018), The Book of Why: The New Science of Cause and Effect, Basic Books.
2. N. BOSTROM (2014), The Ethics of Artificial Intelligence. The Cambridge Handbook of Artificial Intelligence, Cambridge University Press.
3. Proceedings of IJCAI: Workshop on Explainable Artificial Intelligence (XAI), 2019

IOEXXX Pattern Recognition [3-0-0-3]

Course Objectives

- Understand basic concepts in pattern recognition.
- Gain knowledge about state-of-the-art algorithms used in pattern recognition research.
- Understand pattern recognition theories, such as Bayes classifier, linear discriminant analysis.
- Apply pattern recognition techniques in practical problems.

Course Outcomes

- Summarize the various techniques involved in pattern recognition

- Categorize the various pattern recognition techniques into supervised and unsupervised.
- Describe the principles of parametric and non-parametric classification methods.
- Identify the suitable pattern recognition techniques for the particular applications.

Syllabus

Introduction to Pattern Recognition: Basics of Probability, Random Processes and Linear Algebra, Bayes Theorem: Bayes Decision Theory: Minimum-error-rate classification, Classifiers, Discriminant functions, Decision surfaces, Normal density and discriminant functions.

Parameter Estimation Methods: Maximum-Likelihood estimation: Gaussian case; Maximum a Posteriori estimation; Bayesian estimation: Gaussian case. Linear and Non-Linear Classifiers: Perceptron Algorithm - Least-Squares Methods-Nonlinear Classifiers: Multilayer Perceptron's- Back Propagation Algorithm - Combinations of Classifiers – Boosting.

Unsupervised learning: Criterion functions for clustering; Algorithms for clustering: K-Means, Hierarchical and other methods; Cluster validation; Gaussian mixture models; Expectation-Maximization method for parameter estimation; Maximum entropy estimation.

Sequential Pattern Recognition. Hidden Markov Models (HMMs). Discrete HMMs. Continuous HMMs. Nonparametric techniques for density estimation. Parzen-window method. K-Nearest Neighbour method.

Dimensionality reduction - Fisher discriminant analysis; Principal component analysis; Factor Analysis - Linear discriminant functions: Gradient descent; Perceptron; Support vector machines.

Textbooks/References

1. R.O.Duda, P.E.Hart and D.G.Stork, Pattern Classification, John Wiley, 2001
2. S.Theodoridis and K.Koutroumbas, Pattern Recognition, 4th Ed., Academic Press, 2009
3. C.M.Bishop, Pattern Recognition and Machine Learning, Springer, 2006
4. Earl Gose, Richard Johnsonbaugh and Steve Jost, "Pattern Recognition and Image Analysis", Prentice Hall of India, 2002.
5. P.A.Devijer & J.Kittler, "Pattern Recognition-A Statistical Approach", Prentice – Hall, 1990.
6. R.Schalkoff, "Pattern Recognition –Statistical, Structural and Neural Approaches", John Wiley, 1992

IOEXXX Deep Learning [3-0-0-3]

Course Prerequisite

Machine Learning

Course Objectives

- The objective of this course is to cover the fundamentals of neural networks as well as some advanced topics such as recurrent neural networks, long short-term memory cells and convolutional neural networks.
- The objective of this course is to cover the fundamentals of deep learning.
- The course helps to understand the working of Convolutional Neural Networks, RNN and attention models in decision making.
- The course also requires students to implement programming assignments related to these topics.

Course Outcomes

- Identify the deep learning algorithms which are more appropriate for various types of learning tasks in various domains.
- Implement deep learning algorithms and solve real-world problems.
- Implement TensorFlow-based deep learning solutions to classification problems

Syllabus

Introduction: Biological Neuron, Idea of computational units, McCulloch-Pitts unit and Thresholding logic, Linear Perceptron, Perceptron Learning Algorithm, Linear separability. Convergence theorem for Perceptron Learning Algorithm.

Artificial Neural Networks: Review of artificial neurons, single and multi layer perceptrons, perceptron learning algorithm, its convergence proof, different activation functions

Review of Feedforward Networks: Multilayer Perceptron, Gradient Descent, Backpropagation, Empirical Risk Minimization, regularization.

Deep Neural Networks: Difficulty of training deep neural networks, Greedy layerwise training.

Better Training of Neural Networks: Newer optimization methods for neural networks (Adagrad, adadelta, rmsprop, adam, NAG), second order methods for training, Saddle point problem in neural networks, Regularization methods (dropout, drop connect, batch normalization).

Convolutional Neural Networks: Architectures, convolution / pooling layers, LeNet, AlexNet.

Recurrent Neural Networks: Back propagation through time, Long Short Term Memory, Gated Recurrent Units, Bidirectional LSTMs, Bidirectional RNNs.

Generative models: Restrictive Boltzmann Machines (RBMs), Introduction to MCMC and Gibbs Sampling, gradient computations in RBMs, Deep Boltzmann Machines.

Deep Unsupervised Learning and Recent Trends: Autoencoders (standard, sparse, denoising, contractive, etc), Variational

Autoencoders, Adversarial Generative Adversarial Networks, Autoencoder and DBM, Multi- task Deep Learning, Multi-view Deep Learning.

Applications of Deep Learning to Computer Vision: Image segmentation, object detection, automatic image captioning, Image generation with Generative adversarial networks, video to text with LSTM models. Attention models for computer vision tasks.

Applications of Deep Learning to NLP: Introduction to NLP and Vector Space Model of Semantics Word Vector Representations: Continuous Skip-Gram Model, Continuous Bag-of-Words model (CBOW), Glove, Evaluations and Applications in word similarity, analogy reasoning. Introduction to Transformers.

Textbooks/References

1. Ian Goodfellow and Yoshua Bengio and Aaron Courville, Deep Learning, MIT Press, 2016.
2. Bishop, C. ,M., Pattern Recognition and Machine Learning, Springer, 2006
3. Raúl Rojas, Neural Networks : A Systematic Introduction, Springer, 1996

SEMESTER VIII

CSS422 Applied Predictive Analytics [3-0-0-3]

Course Objectives

- To extend student's knowledge in the area of Data Science with emphasis on Predictions utilizing associated statistical methods and software tools.
- Students shall better understand the use in its commercial application to a domain such as Telecommunication/ Insurance and should be able to independently work on a dataset using RStudio.

Course Outcomes

- Ability to apply specific statistical and regression analysis methods applicable to predictive analytics to identify new trends and patterns, uncover relationships, create forecasts, predict likelihoods, and test predictive hypotheses.
- Ability to develop and use various quantitative and classification predictive models based on various regression and decision tree methods.
- Develop familiarity with popular tools and software used in industry for predictive analytics, especially R, R Studio.
- Learn how to apply the models to interpret and report on results for a management audience for a domain such as Telecommunications / Insurance.

Syllabus

Introduction to supervised, unsupervised and semi-supervised modelling / learning.

Regression model building framework: Problem definition, Data pre-processing; Model building; Diagnostics and validation.

Simple Linear Regression: Coefficient of determination, Significance tests, Residual analysis, Confidence and Prediction intervals.

Multiple Linear Regression: Multiple coefficient of determination, Interpretation of regression coefficients, Categorical variables, Heteroscedasticity, Multi-collinearity, outliers, Auto regression and transformation of variables, Regression model building.

Logistic and Multinomial Regression: Logistic function, Estimation of probability using logistic regression, Deviance, Wald Test, Hosmer Lemshow Test, Classification table, Gini co-efficient; Classification and Regression Tree (CART), Decision Tree modelling.

Textbooks/References

1. James, Witten, Hastie and Tibshirani "An Introduction to Statistical Learning: with Applications in R" by, Springer, 1st. Edition, 2013.

2. Alberto Cordoba, “Understanding the Predictive Analytics Lifecycle”, Wiley, 2014.
3. Eric Siegel, Thomas H. Davenport, “Predictive Analytics: The Power to Predict Who Will Click, Buy, Lie, or Die”, Wiley, 2013.
4. James R Evans, “Business Analytics – Methods, Models and Decisions”, Pearson 2013.
5. R. N. Prasad, Seema Acharya, “Fundamentals of Business Analytics”, Wiley, 2015.

CSSXXX Edge Computing [3-0-0-3]

Course Prerequisites

The participants should have prior knowledge on the following topics/courses:

- Computer Networks
- Cloud Computing
- Operating Systems
- Machine Learning

Course Objectives

The objectives of the course are

- To introduce the concept and the basics of edge computing.
- To provide knowledge on various applications of edge-assisted applications.
- To enable awareness on the different edge accelerators.

Course Outcomes

The outcomes of the course are

- The students will have knowledge on various edge-assisted applications.
- The students will have knowledge on the available edge-assistive frameworks or orchestration platforms.
- The students will be capable to design their own applications that utilize edge/fog computing systems.

Syllabus

Introduction – Basics of Computing techniques, Edge computing, Objectives of edge computing, Edge-Fog-Cloud computing requirements, Edge computing challenges.

Edge Computing Architectures – IoT basics, IoT-enabled components, Edge-level microcontrollers and microprocessors, Edge-AI architectures – Generic, Layered, SoA, Interoperability, Metric-oriented architectures.

Edge-level Programming – Edge OS concepts, FreeRTOS, process stages in edge-OS, service-based programming, node-red assisted programming, cloud-assisted edge platforms – AWS Greengrass platforms.

Learning/Inferencing – ML and DL basics, Federated Learning, DNN Splitting, Transfer learning, Gossip Learning, Distributed inferences

Frameworks and Applications – Resource allocation frameworks, Cloud-specific frameworks, Application-specific frameworks, Distributed FL frameworks, Orchestration frameworks, Edge-AI

applications – for healthcare, for agriculture, for industry.

Textbooks/References

1. Shajulin Benedict, Edge Intelligence: Deep Learning-Enabled Edge Computing, in IOP Publishers, UK (London), Hardback ISBN: 9780750355919, Ebook ISBN: 9780750355933, DOI: 10.1088/978-0-7503-5593-3, June 2024.
2. K. Anitha Kumari, G. Sudha Sadasivam, D. Dharani, M. Niranjanamurthy, Edge Computing Fundamentals, Advances and Applications, 2021 CRC Press.
3. Michael Harvey, Intelligence at the Edge, SAS Institute publisher, ISBN: 9781642957785, 2020.
4. Yu Chen and Ronghua Xu, Edge computing architecture and applications for smart cities, Intech Open publisher, 2024.
5. Sean Howard, Edge Computing with AWS: A practical Guide to architecting Secure Edge Cloud infrastructure with AWS.

CSSXXX Computer Systems for Machine Learning [3-0-0-3]

Course Prerequisites

- Machine Learning
- Computer Architecture

Course Objectives

- To understand the basic requirements and typical workloads for machine learning.
- To understand various emerging computer architecture for machine learning.
- To evaluate recent implementation trends and opportunities in machine learning systems.

Course Outcomes

- Understand the key design considerations for machine learning systems.
- Be able to evaluate different considerations for machine learning systems.
- Understand the strengths and weakness of various design techniques for efficient processing for machine learning.

Syllabus

Introduction, Software 2.0, Role of hardware accelerators in post Dennard and Moore era; Classical ML algorithms, Regression, SVMs.

Linear algebra fundamentals and accelerating linear algebra, BLAS operations, 20th century techniques, Systolic arrays and MIMDs, CGRAs, Deep Learning and Tensorflow Basics, Density, Parallelism, and Values, Linear-algebra Processor and Neuflow.

Memory Hierarchies: Diannao, Eyeriss 1 and 2, Quantization and Pruning, Compression of Recurrent DNN.

More Complexity: Attention and Transformers, Industry Inference Products (TPU, Cambricon).

Introduction to Spatial: Analyzing Performance and Energy with Spatial; MLPs and CNNs inference; Evaluating performance, Energy efficiency, Parallelism, Locality, Memory hierarchy, Roofline model, Design Space Optimization with Spatial.

Architecture: Predictive Activation, Temporal Similarity, Smart Training: Lottery Tickets and Hash-based, FPGA-based acceleration: BrainWave and FPGA Limitations, Architecture: Spatial Similarity: PRA and Diffy, Industry - GPU's (Volta) and Architecture: Neural Cache.

Beyond Digital: Flash and Memristor based inference, DNN Accelerators and Conventional Computer Architecture.

Textbooks/References

1. Deep Learning Book, Ian Goodfellow, Yoshua Bengio, Aaron Courville.
2. Machine Learning, Andrew Ng, Coursera.
3. Neural Networks for Machine Learning, Geoffrey Hinton, Coursera.
4. Tensorflow, GPU resources, NVIDIA Deep Learning Accelerator <http://nvidia.org/>
5. List of commercial AI Chips and Projects
6. 2017 UChicago System seminar on Computer Architectures for Machine Learning.

CSSXXX Scientific Machine Learning [3-0-0-3]

Course Objectives

- To provide an introduction to the algorithmic and computational foundations of Scientific Machine Learning
- To provide a basics of scientific computation algorithms.

Course Outcomes

- Understand the algorithmic and computational foundations of Scientific Machine Learning
- Understand basic SciML algorithms: automatic differentiation, physics-informed neural networks, and physics-informed Gaussian processes.

Syllabus

Introduction- An overview of scientific machine learning and scientific laws –Equations-Examples. Review of Machine Learning, Overview of least-squares parametric regression using linear and neural network models, Non-parametric regression using Gaussian processes.

Introduction to Scientific Computing- Basic notions of differential equations, Numerical solutions using classical discretization schemes, Partial Differential Equations.

Automatic Differentiation-Introduction to forward-mode, reverse-mode automatic differentiation with examples, Exposition of back propagation training for neural networks.

Forward Modelling with Physics-Informed Neural Networks (PINNs) - Description of the PINN algorithm, PINN training: Initialization, gradient descent with adaptive learning rate and momentum, learning rate schedule, second-order methods, Self-Adaptive PINNs.

Forward Modelling with Physics-Informed Gaussian Processes (PIGPs), Introduction to Reproducible Kernel Hilbert Spaces (RKHS), Solving nonlinear PDEs by Gaussian Processes as optimal recovery in a RKHS.

Textbooks/References

1. Baker et al.(2019) Workshop report on Basic Research needs for scientific machine learning: Core Technologies for Artificial Intelligence <https://doi.org/10.2172/1478744>
2. Yifan et al. (2021) Solving and learning nonlinear PDEs with Gaussian processes, Journal of Computational Physics, Volume 447.
3. M. Raissi et al. (2019) Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations, Journal of Computational Physics, Volume 378.
4. A. Iserles (2009) A first course in the numerical analysis of differential equations, Cambridge University Press.
5. John P. Boyd (2013) Chebyshev and Fourier Spectral Methods, Second Revised Edition, Dover Publications.
6. Martin et al. (2006) Automatic Differentiation: Applications, Theory, and Implementations, Springer Berlin, Heidelberg.

CSSXXX Time Series Analytics [3-0-0-3]

Course Prerequisite

- Probability, Statistics.

Course Objectives

- To provide working knowledge of time series and forecasting methods.
- To provide with techniques and receipts for estimation and assessment of quality of economic models with time series data.

Course Outcomes

- Gain a working knowledge of time series and forecasting methods.
- Develop the basic skills needed to do empirical research.

- Do empirical research in fields operating with time series data set.
- Get working knowledge of time series and forecasting methods.

Syllabus

Introduction— Time series, time series analysis, and time series forecasting. Stationary Models and the Autocorrelation Function, Estimation and Elimination of Trend and Seasonal Components.

Overview of Time Series Forecasting, Flavors of Machine Learning for Time Series Forecasting, Supervised Learning for Time Series Forecasting. Time Series Forecasting Template-Business Understanding and Performance Metrics-Data Ingestion, Data Pre-processing and Feature Engineering, An Overview of Demand Forecasting Modeling Techniques.

Time Series Data Preparation, Common Data Preparation Operations for Time Series. Time stamps vs. Periods, Time Series Exploration and Understanding, Time Series Feature Engineering.

Introduction to Autoregressive and Automated Methods for Time Series Forecasting. Introduction to Neural Networks for Time Series Forecasting.

Textbooks/References

1. Francesca Lazzeri PhD (2020), Machine Learning for Time Series Forecasting with Python, John Wiley & Sons, Inc. DOI:10.1002/9781119682394
2. P.J. BLOCKWELL, R.A. DAVIS (2017), Introduction to Time Series and Forecasting, Springer, 2nd Edition.
3. C. CHATFIELD (2004), The Analysis of Time Series – An Introduction, Chapman and Hall / CRC, 4th Edition.
4. G.E.P. BOX, G. JENKINS, G. REINSEL (1994), Time Series Analysis-Forecasting and Control, Pearson, 3rd Edition 2020.

CSSXXX Graph Analytics for Big Data [3-0-2-4]

Course Prerequisites

- CSE 412 Big Data Analytics
- IMA 111 Discrete Mathematics (Graph Theory)

Course Objectives

- To model, store, retrieve, graph-structured data by applying a combination of statistical, graph theoretic, and graph database techniques.
- To demonstrate a problem into a graph database and perform analytical tasks over the graph in a scalable manner.

Course Outcomes

- Modelling, storing, retrieving and analysing graph-structured data.

- Designing graph algorithms in various graph analytics systems, and how to choose the most appropriate system for a particular application.
- Computational models of various graph processing systems.
- Effective graph visualizations in order to enhance human perception.

Syllabus

Introduction to Big Graph (Graph + Big Data): Introduction to Big Graph, Scalability of Big Graph: How Big Is Big Graph?- Impact of V's of Big Graph - Graph Databases - Issues and Challenges of Big Graph Analytics (High-Degree Vertex, Sparseness, Data-Driven Computations, Unstructured Problems, In-Memory Challenge, Communication Overhead, Load Balancing), Applications of Big Graph Analytics in various domains.

Big Graph Analytic Techniques: Centrality Analysis (Degree Centrality, Eigenvector Centrality, Katz Centrality, PageRank Centrality, Closeness Centrality, Betweenness Centrality) Path Analysis (Community Analysis, Connectivity Analysis).

Algorithms for Big Graph Analytics: PageRank, Connected Component, Distributed Minimum Spanning Tree, Graph Search Systems for large-scale graph computation: Need for large scale graph computation - Big Graph Analytics Approaches - In-Memory Big Graph Analytics (GraphX) – SSD Based Big Graph Analytics - Disk-Based Big Graph Analytics– Graph Database (Neo4j).

Large Graph Visualization: Need for visualise Big Graph - Issues and challenges in visualising Big Graphs – Layouts (Force-directed and Energy-Based, Dimension Reduction, Node Features Based), Large Graph Visualization tools (GraphViz, Gephi).

Case Study on Behaviour Analytics- Social Network Analysis- Insurance Fraud detection.

Lab Practice

- Install Neo4J in standalone computers and Big data clusters
- Creating Nodes and Relationships in Neo4j
- Querying and Indexing in Neo4J
- Implementing Graph Algorithms in Neo4J
- Install Spark GraphX in standalone computers and Big data clusters
- Spark GraphX Libraries
- Perform Graph operations on GraphX
- GraphX Analytics (Page rank)
- Case Study: Flight data Analysis using Spark GraphX

Textbooks/References

1. Needham, M., & Hodler, A. E. (2019). Graph algorithms: practical examples in Apache Spark and Neo4j. O'Reilly Media.
2. Singh, D. K., Pramanik, P. K. D., & Choudhury, P. (2018). Big graph analytics: techniques, tools, challenges, and applications. Data Analytics, 171-197.
3. Yan, D., Tian, Y., & Cheng, J. (2017). Systems for Big Graph Analytics. Springer International Publishing.
4. Brath, R., & Jonker, D. (2015). Graph analysis and visualization: discovering business opportunity in linked data. John Wiley & Sons.
5. Vukotic, A., Watt, N., Abedrabbo, T., Fox, D., & Partner, J. (2015). Neo4j in action (Vol. 22). Shelter Island: Manning.
6. Malak, M., & East, R. (2016). Spark GraphX in action. Simon and Schuster.
7. Ramamonjison, R. (2015). Apache Spark Graph Processing. Packt Publishing Ltd.

CSSXXX Machine Learning Operations (MLOps) [3-0-2-4]

Course Prerequisite

- Machine Learning
- Introduction to DevOps and Microservices

Course Objectives

- To divulge knowledge on how to scale machine learning in the enterprise
- To provide comprehension of various activities involved in the development, deployment, and monitoring of ML models
- To disseminate the basic principles of MLOps

Course Outcomes

- Understand the novel techniques on how to scale machine learning in the enterprise
- Identify key metrics to optimize model performance
- Understand the various deployment options

Syllabus

Introduction, Defining MLOps and Its Challenges, Difference between MLOps and DevOps. MLOps to Mitigate Risk. MLOps for Responsible AI-two dimensions. Roles of people in the machine learning life cycle.

Key MLOps Features-Model Development, Productionalization and Deployment, Monitoring, iteration and Governance, Developing Models. Preparing for Production- Runtime Environments, Model Risk Evaluation and Mitigation, Machine Learning Security.

Deploying to Production- CI/CD Pipelines, Building ML

Artifacts, Deployment Strategies, Scaling Deployments.

Monitoring and Feedback Loop.- Understanding Model Degradation, Drift Detection, Model evaluation.

Model Governance-Matching Governance with Risk Level, The New Wave of AI-Specific Regulations-Key Elements of Responsible AI, A Template for MLOps Governance, MLOps: Real-World Examples.

Lab Practice

- Machine Learning Systems: Concepts and Stages (data collection to model development). MLOps - Framework and Software Stack MLOps fundamentals of continuous Integration & Continuous Delivery (CI/CD). Simple MLOps implementation for automating model, training and deployment.

Textbooks/References

1. Treveil, Mark, Nicolas Omont, Clément Stenac, Kenji Lefevre, Du Phan, Joachim Zentici, Adrien Lavoillotte, Makoto Miyazaki, and Lynn Heidmann. Introducing MLOps. O'Reilly Media, 2020.
2. Burkov, Andriy. Machine Learning Engineering. True Positive Inc. , 2020.
3. Ameisen, Emmanuel. Building Machine Learning Powered Applications. O'Reilly Media, 2020.
4. <https://stanford-cs329s.github.io/>
5. <https://ucbrise.github.io/cs294-ai-sys-sp19/>
6. <https://ckaestne.github.io/seai/>
7. <https://www.coursera.org/specializations/machine-learning-engineering-for-production-mlops#courses>

IOE421 Large Language Models [2-0-2-3]

Course Prerequisite

- Probability and Statistics
- Machine Learning

Course Objectives

- Understand foundational concepts in natural language processing methods and strategies
- Provide a comprehensive understanding of transformer models, pre-training, and fine-tuning for modern Large Language Models (LLMs).
- Apply Transformers and Large Language models for building various NLP applications
- Develop awareness of the ethical implications, biases, and challenges in deploying responsible and fair LLM systems.

Course Objectives

- Describe the core NLP tasks, challenges, and the evolution from traditional methods to neural approaches in NLP.
- Analyze the architecture and operational principles of transformer-based models
- Apply pre-trained LLMs for practical tasks such as text classification, summarization, and text generation
- Evaluate the ethical and societal implications of deploying LLMs, focusing on bias, fairness, and responsible AI practices.

Syllabus

Introduction to Natural Language Processing: overview of NLP tasks, challenges, evolution. Review of Deep Neural networks-feedforward network, Backpropagation, RNN, Neural Language Models: Word Embedding. Seq2seq models: Encoder-decoder, Attention mechanism

Transformers and Large Language Models: Tokens and Embedding, Transformer Blocks, Self attention, Masked attention, Encoder only models (BERT), Decoder only models (GPT), Pre-training and Fine tuning , Applications of pre-trained models for Text classification, Summarization, Text generation

LLM Techniques and Applications: Decoding strategies, Prompt Engineering: In-Context Learning, and Instruct Tuning, Retrieval augmented generation, Alignment and PEFT, Responsible and Ethical LLMs: Bias and fairness, Guardrails and jailbreaking,

Lab Practice

- Basics of text processing: Processing Raw Text
- Text classification
- Sentiment Analysis
- Word embedding
- Transformers using huggingface/Langchain
- Fine Tuning a transformer model
- Prompt Engineering
- Retrieval Augmented Generation

Textbooks/References

1. Alammam, Jay, and Maarten Grootendorst. Hands-On Large Language Models: Language Understanding and Generation. O'Reilly Media, Inc., 2024.
2. Jurafsky, Daniel, and James H. Martin. "Speech and Language Processing: An Introduction to Natural Language Processing, Computational Linguistics, and Speech Recognition.", 3rd Edition, 2024.
3. Amaratunga, Thimira. "Understanding large language models: learning their underlying concepts and technologies." , Apress Media, 2023.
4. Vaswani, A. "Attention is all you need." Advances in Neural Information Processing Systems (2017).

BOUQUET CORE COURSES

CSEXXX Advanced Computer Networks

Course Objectives

- The students with the advanced concepts of computer networks.
- The students will be introduced to the basics of networking; will be introduced to WSNs; and,
- A comparative study of wired and wireless networks.
- Practical sessions covering wired and wireless networking concepts via. simulations (for instance, Network Simulator-3).

Course Outcomes

- Understood the concepts of wired/wireless networks (WSNs),
- Learned the importance of the energy concerns of WSNs,
- Compared the protocols of wired/wireless networks;
- Simulated the topologies of wired/wireless networks; and,
- Studied the application of networks in emerging ICT domains.

Syllabus

Basics of computer networks –Types of networks, Topologies, Conversions, Data transmission basics, Bandwidth utilization techniques, Spread spectrum techniques, Layers and standards.

Wired/Wireless Phy/DLL – Introduction to WSN, WSN-Single node architecture, Sensors and actuators, Design principles of WSN, Communication protocols – phy layer, MAC protocols (Wired and wireless).

Wired/Wireless DLL/NL – Naming and addressing, Topology control, routing protocols for wired / wireless networks, Topology design using NS3 Toolkit.

Advanced Topics – Data centric networking, content based networking, Data center networking, Networking and virtualization, Software Defined Networking, Cloud networking.

Lab Practice

- NS3 and Eclipse IDE installation.
- Logging exercises of NS3.
- Topology study using NS3.
- Tracing using NS3.
- Public/private key authentication.
- Passwordless connection establishment.
- Setting up firewall.

Textbooks/References

1. Andrew S. Tanenbaum, Computer Networks, Pearson Education, Fifth Edition, 2003.
2. Fred Halsal, Data communications Computer Networks and Open Systems, Pearson Education, Fourth Edition, 1992.
3. Holger Karl and Andreas Willing, Protocols and Architectures for Wireless Sensor Networks, John Wiley and Sons, First Edition, 2005.
4. Paul Gorransson and Chuck Black, Software Defined Networks - A Comprehensive Approach, Morgan Kauffman publishers, First Edition, 2014.

CS ELECTIVE COURSES

CSSXXX Parallel Algorithms

Course Objectives

- Learn principles and design techniques of parallel algorithms, parallel and distributed algorithms development techniques for shared memory and message passing models.
- To study the main classes of parallel algorithms.
- To study the implementation of parallel algorithms using PyPVM, PyMPI and openmp. This course is aimed at undergraduate students who are interested in the algorithmic aspect of parallel computation.

Course Outcomes

- Understand fundamental concepts of parallel algorithmics
- Design parallel algorithms for sorting, searching, matrix algorithms etc
- Write parallel programs in PyPVM and PyMPI
- Understand Fourier transforms and its parallel versions
- Understand connection networks like butterfly, omega and other networks

Syllabus

History of computers, Examples of parallel computers, Application areas of supercomputers, Von Neumann architecture, Flynn's Taxonomy (SISD, SIMD, MISD, MIMD), SPMD, Vector/Array and Grid processors, Cluster computers, Master-slave vs Client-server models, vertical vs horizontal scaling of CPU, Map-Reduce models.

Performance measures, Speed-up, Efficiency, Gantt chart, Amdahl's law, Gustafson-Barsis law, granularity and scalability.

Shared memory architectures: PRAM models, CRCW, CREW, ERCW, EREW, NUMA etc Memory interconnections – BUS, CROSSBAR, OMEGA NETWORKS, TREE, PERFECT SHUFFLE Threads vs processes, issues in thread-based parallel algorithms. Some Shared memory algorithms: sieve of Erasthosthenes, LU Decomposition, Odd-Even Transposition sort, Batcher's Bitonic Sort.

Distributed Memory Machines :Examples of computers in this category, and software packages PVM, MPI, OpenMP Basics of message passing, process creation, synchronisation, some application areas Hypercube architectures.

Communication Methods-peer-to-peer, collective broadcast, one-to-all, all-to-all, gather and scatter, other.

Criteria for model selection: Communication cost and Latency, Load Balancing, Processor scheduling, Scalability, Fault-Tolerance.

Load balancing:Dynamic load balancing - synchronized vs asynchronized, centralized vs decentralized, termination detection algorithms, graph representations.

Divide and Conquer (D&C) algorithms:D&C vs decrease and conquer, parallel implementation of D&C algorithms, partitioning strategies.

Matrix algorithms:Transposition, LU decomposition, determinant, solving linear equations, matrix-vector multiplication, matrix multiplication, Strassen's algorithm, matrix inverse.

Sorting algorithms:Bubble sort, insertion sort, merge sort, quick sort, odd-even transposition sort, Batcher's bitonic sort, bucket-sort, sorting on hypercube, sorting networks Searching algorithms.

Linear search, binary search, parallel searching strategies, speedup anomalies in Parallel searching algorithms, Graph algorithms.

BFS, DFS, spanning trees, Shortest path, Dijkstra's and Prims algorithms. Fast Fourier Transform, Serial algorithm, Binary-Exchange algorithm, Transpose algorithm.

Emerging areas of parallel algorithms.

Textbooks/References

1. Parallel Programming: Techniques and Applications Using Networked Workstations and Parallel Computers, 2nd Edition By Barry Wilkinson, Michael Allen Pearson; 2 edition (4 March 2004).
2. An Introduction to Parallel Computing: Design and Analysis of Algorithms, 2e Paperback 2004 Anand Grama, Anshul Gupta, George Karypis, Vipin Kumar Pearson Education India; 2 edition (2004).
3. Parallel Computing: Theory and Practice Paperback – 1 Jul 2017 Michael Quinn J, McGraw Hill Education; 2 edition (1 July 2017).
4. Parallel Sorting Algorithms Paperback – 1 Jan 1985 Selim G. Akl, Academic Press (1 January 1985).
5. Parallel Computation: Models and Methods, Selim G. Akl, Pearson (6 November 1996).

CSSXXX Fundamentals of Computer Forensics

Course Prerequisite

- Student should have a passing Grade in ICS111 (IT Workshop I) and ICS 212 (Operating Systems) or the instructor's approval. If a student has not completed the prerequisite or does not meet the prerequisite requirement, he/she cannot enroll in the course. All prerequisites are enforced.

Course Objectives

- Summarize the basic principles of computer forensics.
- Understand various stages of digital forensics investigation.
- Summarize important laws relevant to digital forensics investigation.
- Describe the digital forensic lab and tools.
- Understand storage management and its importance in forensics
- Understand the file system fundamentals and the internals of Windows and Linux file systems.

Course Outcomes

- Understand the principles and stages of computer forensics.
- Comprehend various legal aspect pertaining to digital forensic investigation.
- Understand various Windows and Linux Files systems and its operations.
- Grasp digital forensic tools and the setup of a digital forensic lab.

Syllabus

Introduction to Computer Forensics, Foundations of Digital Forensics, Need for Digital Forensics, Forensic Investigation Model.

Overview of computer hardware and operating systems, Storage media/devices and their working, Windows and Linux File Systems.

First Responder, Forensic data acquisition, Forensic data validation, Evidence Collection and Analysis Techniques, Digital Forensics Lab and Forensic tools.

Computer crime and Legal issues: IT Act of 2000, Data Protection Act in European Union.

Textbooks/References

1. Bill Nelson, Amelia Phillips, Christopher Steuart, Guide to Computer Forensics and Investigations, 6th Edition, Publisher: Cengage Learning.
2. Chuck Easttom, System Forensics Investigation and Response, 3rd Edition, Publisher: Jones & Bartlett Learning.

3. Cory Altheide, Bruce Nikkel, Harlan Carvey, Digital Forensics with Open Source Tools, Publisher: Elsevier publication.
4. Marjie T. Britz, Computer Forensics and Cyber Crime: An Introduction, Publisher: Pearson.
5. Cory Altheide, Bruce Nikkel, Harlan Carvey, Digital Forensics with Open Source Tools, Publisher: Elsevier Science & Technology.

CSSXXX Advanced Network Security

Course Prerequisite

- ICS224 Computer Networks

Course Objectives

- Comprehend the necessity of network security along with the basic concept of Network security.
- Analyse passwords and understand the security issues like Brute force attack, Dictionary attack and rainbow table attack associated with passwords.
- Investigate various network vulnerabilities like virus, worm, malware, rootkit and devise strategies to mitigate them.
- Analyse privacy threatening behaviour over the internet and formulate defensive techniques to preserve privacy.
- Discuss and learn data privacy laws in India and in Europe. Examine Information Technology Act in India along with its amendments.

Course Outcomes

- Understand the need for network security and have through grasp of the fundamentals of network security.
- Recognise network vulnerabilities and develop Network defensive strategies by utilizing Intrusion Detection Systems, Honeypot etc.
- Comprehend various legal perspective regarding data security and privacy in India and Europe. Develop a clear perception of Information Technology Act in India.
- Identify and defend against various privacy threatening tools and techniques over the internet.

Syllabus

Introduction to Network Security, Need for Network Security, Network Security Fundamentals, Principles of Security, DNS Vulnerabilities, Secure Network Communication.

Access Control, Password Attacks, Password Policies, Rainbow Table Attack, Social Engineering, Malware, Insider Attack and Defence, Computer Virus Types and Defence, Computer Worms, Rootkits, Botnet, Denial of Service Attack.

Need For Physical Security, User Authentication Technologies, Environmental Attacks and Accidents, Special-Purpose Machines and Security, Firewall, Intrusion Detection System, Honeypot, Tunnelling, Virtual Private Network.

Privacy Preserving Communication, Anonymity, Onion Routing, Device Fingerprinting, Data Protection Laws in India, Data Protection Act in European Union, Information Technology Act of 2000 and its amendments.

Textbooks/References

1. Michael Goodrich, Roberto Tamassia, Introduction to Computer Security: Pearson New International Edition, Published 15 Aug 2013, ISBN-10: 1292025409, ISBN-13: 9781292025407.
2. Kun Peng, Anonymous Communication Networks: Protecting Privacy on the Web, Auerbach Publications, 1st edition, Published: 7 April 2014, ISBN-10: 143988157X, ISBN-13: 978-1439881576.
3. Christopher Hadnagy, Social Engineering: The Science of Human Hacking, Publisher: Wiley, 2nd Edition Publication date: July 31, 2018, ISBN-10: 111943338X, ISBN-13: 978-1119433385.
4. Chris Sanders, Jason Smith, Applied Network Security Monitoring: Collection, Detection, and Analysis, Publisher: Syngress, 1st edition Publication Date: December 19, 2013, ISBN-10: 0124172083, ISBN-13: 978-0124172081.
5. Sagar Rahalkar, Quick Start Guide to Penetration Testing: With NMAP, OpenVAS and Metasploit, Publisher: Apress; 1st edition Publication Date: 21 Jan. 2019, ISBN-10: 1484242696, ISBN-13: 978-1484242698.

Research Papers

1. A. C. Jose, R. Malekian, and N. Ye, "Improving home automation security; Integrating device fingerprinting into smart home," IEEE Access, vol. 4, pp. 5776–5787, 2016.
2. J. R. Mayer, "Any person... a pamphleteer: Internet anonymity in the age of Web2.0," M.S.thesis, Dept. School Public Int. Affairs, Princeton Univ., Princeton, NJ, USA, 2009.
3. P. Eckersley, "How Unique Is Your Browser?" in Proc. 10th Privacy Enhancing Technol. Symp. (PETS), Berlin, Germany, Jul. 2010, pp. 1–35.
4. T.-F. Yen, Y. Xie, F. Yu, R. P. Yu, and M. Abadi, "Host fingerprinting and tracking on the Web: Privacy and security implications," in Proc. 19th Annu. Netw. Distrib. Syst. Secur. Symp. (NDSS), San Diego, CA, USA, Feb. 2012.
5. N. Nikiforakis, A. Kapravelos, W. Joosen, C. Kruegel, F. Piessens, and G. Vigna, "Cookieless monster: Exploring the ecosystem of Web-based device fingerprinting," in Proc. IEEE Symp. Secur. Privacy, May 2013, pp. 541–555.
6. K. Mowery, D. Bogenreif, S. Yilek, and H. Shacham, "Fingerprinting information in JavaScript implementations," in Proc. W2SP, May 2011, pp. 1–11.

7. T. Kohno, A. Broido, and K. Claffy, "Remote physical device fingerprinting," IEEE Trans. Dependable Secure Comput., vol. 2, no. 2, pp. 93–108, Apr./Jun. 2005.
8. S. Zander and S. J. Murdoch, "An improved clock-skew measurement technique for revealing hidden services," in Proc. 17th Conf. Secur. Symp., Berkeley, CA, USA, 2008, pp. 211–225.

CSSXXX Human Computer Interaction

Course Prerequisite

- OOAD

Course Objectives

- Learn the foundations of Human Computer Interaction.
- Be familiar with the design technologies for individuals and persons with disabilities.
- Be aware of mobile HCI.
- Learn the guidelines for user interface.

Course Outcomes

Syllabus

Foundation of HCI: HCI foundation and history; The Human: I/O channels – Memory – Reasoning and problem solving; The computer: Devices – Memory – processing and networks; Interaction: Models – frameworks – Ergonomics – styles – elements – interactivity- Paradigms.

Design and Software Process: Interactive Design basics – process – scenarios – navigation – screen design – Iteration and prototyping. HCI in software process – software life cycle – usability engineering – Prototyping in practice – design rationale. Design rules – principles, standards, guidelines, rules. Evaluation Techniques – Universal Design.

Model-based Design and evaluation: Basic idea, introduction to different types of models, GOMS family of models (KLM and CMN-GOMS), Fitts' law and Hick-Hyman's law, Model-based design case studies.

Guidelines in HCI: Shneiderman's eight golden rules, Norman's seven principles, Norman's model of interaction, Nielsen's ten heuristics with example of its use, Heuristic evaluation, Contextual inquiry, Cognitive walkthrough;

Task modelling and analysis: Hierarchical task analysis (HTA), Engineering task models and Concur Task Tree (CTT).

Dialog Design: Introduction to formalism in dialog design, design using FSM (finite state machines), State charts and (classical) Petri Nets in dialog design.

Cognitive models and architecture: Socio-Organizational issues and stake holder requirements – Communication and collaboration models ; Introduction to Cognitive Architecture, Cognitive Architecture types, relevance of Cognitive Architecture in IS design, Model Human Processor (MHP).

OOM- Object Oriented Modeling of User Interface Design; Design Case Studies.

Textbooks/References

1. Dix A., Finlay J., Abowd G. D. and Beale R. Human Computer Interaction, 3rd edition, Pearson Education, 2005.
2. Preece J., Rogers Y., Sharp H., Baniyon D., Holland S. and Carey T. Human Computer Interaction, Addison-Wesley, 1994.
3. B. Shneiderman; Designing the User Interface, Addison Wesley 2000.
4. Brian Fling, 'Mobile Design and Development', First Edition, O'Reilly Media Inc., 2009.
5. Bill Scott and Theresa Neil, "Designing Web Interfaces", First Edition, O'Reilly, 2009.

MATHEMATICS COURSES

ELECTIVE

IMAXXX Elementary Number Theory and Cryptography

Course Objectives

- To learn the basics of Elementary number theory and Cryptography.
- Use their programming skills to attack some classical problems in Number theory like Pells equation by obtaining continued fraction expansion of quadratic irrationals.

Course Outcomes

Syllabus

Divisibility, Greatest Common Divisor, Euclid's algorithm, Linear diophantine equations, prime numbers, fundamental theorem of arithmetic, Linear Congruences, Chinese remainder theorem, Basics of group theory, examples of groups such as $\mathbb{Z}_n$ and $(\mathbb{Z}_n)^*$, Group table, order of group, Lagrange's theorem, Wilson's theorem, Fermat's little theorem, Euler's theorem, Arithmetic functions, Euler Totient function, sum of divisors function, Pillai's arithmetic function, Perfect numbers, primitive roots, Quadratic residues, Legendre symbol, law of quadratic reciprocity, Jacobi symbol, Jacobi Reciprocity Theorem, Continued Fractions, Pells Equation, solution of Pells equation using continued fractions. Fermat's sum of 2 squares theorem.

A Final project involving programming and number theory. Challenges by Fermat related to Pells equation was assigned as projects to write a programming code and to obtain the fundamental solution.

Textbooks/References

1. David M. Burton, Elementary Number Theory, 7th Edition, McGraw Hill, 2012.
2. Neal Koblitz, A Course in Number Theory and Cryptography, Springer, Second Edition, 1994.

3. Ivan Niven, H. S. Zuckerman and H. L. Montgomery, An Introduction to the Theory of Numbers, Wiley, Fifth Edition, 1991.
4. Alexander Stanoyevitch, Introduction to Cryptography with Mathematical Foundations and Computer Implementations, Chapman and Hall/CRC, First Edition, 2010.

IMAXXX Optimization Techniques

Course Objectives

- To explain the basic mathematical concepts of optimization.
- To develop the modelling skills necessary to describe and formulate optimization problems.
- To conduct and interpret the post optimal and sensitivity analysis and explain the primal-dual relationship.
- To provide the skills necessary to solve and interpret optimization problems in engineering.

Course Outcomes

- Apply knowledge of optimization to formulate and solve engineering problems.
- Understand the different methods of optimization and be able to suggest a technique for a specific problem.
- Understand how optimization can be used to solve industrial problems.

Syllabus

Classification and general theory of optimization; Linear programming (LP): formulation and geometric ideas, simplex and revised simplex methods, duality and sensitivity, interior-point methods for LP problems, transportation, assignment, and integer programming problems.

Nonlinear optimization, method of Lagrange multipliers, Karush-Kuhn-Tucker theory, numerical methods for nonlinear optimization, convex optimization, quadratic optimization; Dynamic programming.

Textbooks/References

1. D. G. Luenberger and Y. Ye, Linear and Nonlinear Programming, Third Edition, Springer India, 2008.
2. N. S. Kambo, Mathematical Programming Techniques, East-West Press, 1997.
3. E. K. P. Chong and S. H. Zak, An Introduction to Optimization, Second Edition, Wiley India, 2001.
4. M. S. Bazaraa, H. D. Sherali and C. M. Shetty, Nonlinear Programming Theory and Algorithms, Third Edition, Wiley India, 2006.
5. K. G. Murty, Linear Programming, Wiley, 1983.

IMAXXX Estimation and Detection Theory

Course Objectives

- To study estimation techniques.
- To study detection techniques.
- To estimate parameters for a given application.
- To apply detection techniques for a given application.

Course Outcomes

- Apply detection techniques in complex problem.
- Estimate parameters for a given application.
- Analyze performance of communication systems using various detection/estimation schemes.

Syllabus

Fundamentals of Estimation Theory: Role of estimation in signal processing, unbiased estimation, minimum variance unbiased (MVU) estimators, finding MVU estimators, Cramer-Rao lower bound, linear modelling-examples.

Deterministic Parameter Estimation: Least squares estimation, linear and nonlinear, best linear unbiased estimation (BLUE), examples, finding the BLUE, maximum likelihood estimation (MLE), numerical determination of the MLE.

Random Parameter Estimation: Bayesian philosophy, prior knowledge and estimation, choosing a prior PDF, Bayesian linear model general Bayesian estimator, risk functions, minimum mean square error estimator, maximum a posteriori estimator, performance characterization.

Fundamentals of Detection Theory: Bayesian criterion, binary hypothesis testing, Mary hypothesis testing, minimax criterion, Neyman-Pearson criterion, receiver operating characteristic, composite hypothesis testing, θ -random variable and θ -nonrandom and unknown, sequential detection.

Applications of Detection and Estimation: Applications in communications, system identification, pattern recognition, speech processing, and image processing.

Textbooks/References

1. S.M. Kay, Fundamentals of Statistical Signal Processing: Estimation Theory, Prentice hall, 1993.
2. S.M. Kay, Fundamentals of Statistical Signal Processing: Detection Theory, Prentice hall, 1998.
3. M.Barkat, Signal Detection and Estimation, Artech House, 2nd edition, 2005.
4. H.V.Poor, An Introduction to Signal Detection and Estimation, Springer, 2nd edition, 1998.
5. H.L.Van Trees, Detection, Estimation and Modulation Theory, Part-I, John Wiley, 1968.

IMAXXX Algorithmic Graph Theory

Course Objectives

- To introduce and apply algorithmic paradigms for graph theoretic problems.
- Computability and complexity aspects restricted to graph-theoretic problems shall be explored.
- Special graphs and their structural properties shall also be explored. Handling graphs using Neo4j.

Course Outcomes

- Tools for design and analysis of algorithms for graph-theoretic problems shall be the key outcome of this course.
- Art of proof writing, establishing proof of correctness of algorithms and familiarizing Neo4j shall be the additional outcome.

Syllabus

Introduction to search and spanning tree problems - Variants such as second-best spanning tree, Kth-best spanning tree, Steiner tree. Shortest path problems and variants - Dijkstra's algorithm, data structures - Binomial and Fibonacci heap for running shortest path problems, analysis. Bidirectional Dijkstra and A-star algorithms. Introduction to Neo4j - Graph creation and querying - cypher query language - CREATE, MATCH, WHERE and RETURN clauses.

All pair shortest path algorithms. Maximum Flow problems - Ford-Fulkerson algorithm, Maximum bipartite matching algorithms, push-relabel methods and analysis. Implementation of these algorithms using python and Neo4j. Basic Graph-theoretic concepts, Structural properties of special graph classes (chordal graphs, bipartite graphs, chordal bipartite graphs, planar graphs, etc.) and algorithmic aspects along with sophisticated data structures. The complexity of classical problems such as minimum vertex cover, minimum independent set, maximum clique.

Theory of NP-completeness - Classical NP-hard reductions - Reductions in special graphs -Complexity of Hamiltonian cycle/path in special graphs. Dichotomy results - Micro-level analysis of reductions - Discovery of polynomial-time solvable input instances.

Textbooks/References

1. Thomas H. Cormen. Charles E. Leiserson. Ronald L. Rivest. Clifford Stein. Introduction to Algorithms. Third Edition. The MIT Press, 2009.
2. Ian Robinson, Jim Webber and Emil Eifrem. Graph Databases. Second Edition. O'Reilly Media, Inc., 2015. (online free e-book is available)
3. Martin Charles Golumbic, Algorithmic Graph Theory and Perfect graphs, Annals of Discrete Mathematics, 57, 1980.
4. Andreas Brandstadt, Van Bang Le, Jeremy P. Spinrad, Graph Classes - A Survey, Mono-graph, SIAM, 1999.

HUMANITIES ELECTIVE COURSES

IHSXXX Introduction to Sociology

Course Objectives

- To understand the relationship between Science, Technology, Society and Culture.
- To familiarize with the social processes and dynamics of scientific knowledge and developments in Social Sciences
- To introduce Sociology as a science of society vis-à-vis Science and Technology.
- To know the fundamental social processes and institutions such as socialization, marriage, family, caste, gender etc.
- To discuss some of the important issues and problems in India today

Course Outcomes

- The relationship between Science, Technology, Society and Culture.
- Social processes and dynamics of scientific knowledge and developments in Social Sciences
- Sociology as a science of society vis-à-vis Science and Technology
- The fundamental social processes and institutions of socialization, marriage, family, caste and gender.
- Some of the important issues and problems in India today, such as Communalism, Diversity and Pluralism, Fundamentalism, Secularism, Globalization, Liberalization, Information Communication Revolution etc.

Syllabus

Sociology & Social Sciences, Science, Social Science and Society, Scientific Study of Society, Sociology - Science of Society; ,Approaches & Methods: Macro & Micro - Object & Subject, Objectivist & Interpretivist.

Society, Social Structure & Culture ,Nature & Culture: Science and Society Interface ,Society and social structure, Culture,Socialization, Social Institutions - Marriage, Family, Caste, Class & Gender.

India today - major issues and challenges, Diversity & Pluralism, Secularism & Communalism, Globalization & Consumer Culture, Information Communication Revolution - New media.

Textbooks/References

1. Haralambos, Michael with Robin Heald. Sociology - Themes and Perspectives. New Delhi : Oxford University Press, 1980.
2. Mohanty, Manoranjan, Class, Caste, Gender. New Delhi: Sage Publications, 2004.

3. Perry and Perry, Contemporary Society: An Introduction to Social Science. London: Allyn & Bacon, 2008.
4. Bryman Alan, Social Research Methods, Oxford University Press, Oxford, 2005.
5. Srinivas M.N, Caste in India and Other Essays. Delhi: Asia Publishing House, 1962.
6. Patel, Tulasi. The Family in India: Structure and Practice. New Delhi: Sage, 2005.
7. Knott-Cetina, Karin and Michael Mulkay, Science Observed: Perspectives on the
8. Social Study of Science. London: Sage Publications, 1983.Merton, Robert K, The Sociology of Science. Chicago: Univ. of Chicago Press, 1973.
9. Bijker, W. E. , Social construction of technology', International Encyclopedia of the Social and Behavioural Sciences. N. Smelser and P. Bailes, eds. Amsterdam, 2001.
10. Sarah Franklin, "Science as Culture, Cultures of Science," Annual Reviews of Anthropology 1995.
11. Sharma, K.L, Indian Social Structure and Change. Jaipur: Rawat Publications, 2007.
12. Bruno Latour and Steve Woolgar, Laboratory Life: The Construction of Scientific Facts Princeton NJ: Princeton University Press, 1986.

IHSXXX Taxation and Human Development /Economics

Course Objectives

- The course seeks to sensitize the students to concepts of income, human development, government's role in development, governance functions and taxation.
- It aims at raising the capacity of non- economics students to appreciate the need to understand some of these concepts for good citizenship as well for becoming able technologists.

Syllabus

Taxation and Central Transfers, Introduction to National Income, Gross Domestic Product (GDP), Gross Domestic Product and Gross National Income(GNI),Three Approaches to GDP, System of National Accounts and Production Boundary, Household's non-market Production,Women, Work, and the Economy:, Macroeconomic Gains from Gender Equity, Measuring Government Output, Satellite Account,Tourism Satellite Account for India, GDP to Welfare Measures, Human Development Index ,Purchasing Power Parities, National Income and Quarterly estimates of Gross Domestic Product (mospi.nic.in).

Textbooks/References

1. Ann Chadeau () What Is Households' NonMarket Production Worth? OECD Economic Studies No. 18, Spring 1992.
2. Katrin Elborgh-Woytek et al., (2013) Women, Work, and the Economy: Macroeconomic Gains from Gender Equity, IMF SDN/13/10.

3. Lequiller, F. and D. Blades (2014), Understanding National Accounts: Second Edition, OECD Publishing. (<http://dx.doi.org/10.1787/9789264214637-en>)
4. UNDP, Human Development Report 1990 and 2016.

IHSXXX Introduction to Digital Humanities [1-0-0-1]

Course Description

Digital Humanities (DH) looks at how the study of humanities has been immensely transformed by digital technology. It is an emerging discipline that studies the changing methodologies and pedagogies in humanities study and research in the light of the growing intersection of humanities with digital technology. This approach has a new relevance in the context of COVID 19 pandemic that has led to a knockout blow to the traditional approaches in education. Digital Humanities has a two-way approach; the systematic use of digital resources in humanities and the humanistic analysis of the application of digital resources. The discipline requires a certain amount of acquaintance with computer technology and the course can be handled in collaboration with faculties from Computer Science department.

Course Objectives

- To introduce the basic concepts and tools in Digital Humanities
- To familiarize students with new forms of textuality and digital archiving
- To develop competency in digital writing
- To incorporate digital tools and methods to support their own core studies and to develop practical skills in electronic archiving, processing, editing, etc.

Course Outcomes

- critically discuss digital humanities projects in light of current methods and theoretical approaches to the field
- to apply principles of textual, editorial and communication theory to technical situations, so that they can produce high quality work in a digital environment

Syllabus

A history of the Humanities in an age of digital computing- Transitions in humanities- Development of digital humanities- Beginnings of the intersection of humanities with computing- Ethics and the digital community- Digital Libraries and Archiving- Digital Humanities in India- Significant Digital Humanities Projects - Intersections of Digital Humanities with web designing, software programming, technical writing, Game Studies, etc. Job prospects -Data Manager, Data Designer- Digital Curator-Cultural Designer- Information Architect- Data Scientist-Metadata Analyst- Related disciplines (Medical Humanities, Astrobiology, etc.)

Textbooks/References

1. Exploring Digital Humanities in India: Pedagogies, Practices, and Institutional Possibilities Edited By Maya Dodd and Nidhi Kalra, Routledge India, 2020.

2. A Companion to Digital Humanities, Eds. Schreibman, Siemens, and Unsworth, Blackwell, 2004
3. Debates in the Digital Humanities, Ed. Matthew Gold, University of Minnesota Press, 2012
4. Shillingsburg, Peter. (2006) From Gutenberg to Google. Electronic Representations of Literary Texts. Paperback. Cambridge University Press.

IHSXXX Visual Communication [1-0-0-1]

Course Description

Visual Communication examines the production, reception, and interpretation of the aspects of culture that communicate through visual imagery. These include not only art, but photography, and commercial imagery (including advertisements, popular film, and television). The emergence of visual culture is closely linked to the development of visual technologies.

Course Objectives

- To understand the historical, physiological, psychological, perceptual, and cognitive aspects of visual communication
- To critically analyse the ethical implications of electronic and print media images
- To examine how visual culture is inextricably tied to marketing and consumerism in the modern world
- To explore the possibilities of digital marketing

Course Outcomes

- Build a critical understanding of the relationship between images and their representation
- To express the student's own ideas with creativity, clarity, and critical thinking

Syllabus

Introduction to Visual Communication- the politics of visibility- the virtual 'real' in contemporary culture- Consumption of mass culture- Photography, Cinema, Advertisement, Web content, etc.- The Digital World: The Virtual Reality in Contemporary Culture Theories of Visuality and impact on visual communication.

New perspectives in visual culture- virtual usurps in the internet and cyberspaces- The misuse of visual technologies.

Textbooks/References

1. Lester, Paul Martin. Visual Communication Images with Messages 4th Edition. Belmont, CA: Wadsworth Publishing Company, 2006.
2. Jones, Steve. Virtual Culture: Identity and Communication in Cybersociety. New Delhi: Sage, 2002.

IHSXXX Science Fiction and Graphic Narratives [1-0-0-1]

Course Description

The course looks at the origins of science fiction and graphic narratives to understand how science and technology intersect with the world of fiction. This course is designed for a reader community who will approach these works from a critical perspective.

Course Objectives

- To familiarize students with the history, culture and origins of science fiction and graphic narratives
- To appreciate the technological elements in science fiction and graphic narratives
- To promote the close reading of science fiction and visual storytelling materials

Course Outcomes

- Identify the relationship between science and society
- Write analytically about fictional works

Syllabus

Origins of science fiction- sub-genres and themes Utopia- Dystopia- Time travel- Cybernetics- Issues of gender, ecology, etc. in science fiction.

Origins of graphic narratives- Recent trends Emergence of web comics- Print versus web comics comics and technology.

Textbooks/References

1. Paul Gravett. Graphic Novels. New York, Collins, 2005.
2. Bould, Mark et al (ed) - The Routledge Companion to Science Fiction, Routledge, London & New York, 2009.
3. Lester, Paul Martin. Visual Communication

IHSXXX Professional Communication [1-0-0-1]

Course Description

The course would cover the important aspects of communication in the professional scenario.

Emphasis would be given to key language skills inevitable for a successful professional life.

Course Objectives

- To understand the importance of communication in the professional/corporate world
- To attain fluency in oral communication
- To attain proficiency in technical writing
- To develop positive body language
- To learn the etiquettes to be practiced while participating in seminars and group discussions
- To acquire interview skills

- To practice interpersonal skills for better relations with seniors, juniors, peers and stake holders

Course Outcomes

- Draft effective proposals, circulars, minutes, etc.
- Prepare an attractive resume.
- To apply for a job and participate in selection process.
- Overcome the fear of interview process and be ready to perform at different levels of the selection process.
- To project a good personal image and social etiquette so as to have a positive impact on the building of one's chosen career.

Syllabus

Basics of professional/business communication developing social communication skills- vocabulary in the professional scenario- relevance of non-verbal communication- body language- haptics- chronemics Technical writing- writing proposals- email etiquettes- basics of research writing- abstract preparation- drafting a resume- cover letter- digital profile-video profile.

Interview skills- Group discussion.

Textbooks/References

1. Tebeaux, Elizabeth, and Sam Dragga. The Essentials of Technical Communication. OUP, 2018.
2. Raman, Meenakshi and Sangeeta Sharma. Technical Communication: Principles and Practice. OUP, 2015.

IHSXXX Advanced Communication Skills [1-0-0-1]

Non-Credit Course

Course objectives

- This course is designed to build the vocabulary and language proficiency that are appropriate for IT professionals.
- This course will enable them to recognize problems with listening style and help them use listening skills to meet various professional needs.
- Effective communication abilities are helpful in comprehending, developing, and presenting technical presentations.
- Active listening is necessary to grasp the context in order to solve problems.
- To eliminate uncertainty during team activities, writing abilities must be standardized and sound.

Course outcomes

- Understand and use various listening and speaking etiquette in everyday communication.
- Successfully read, comprehend, and critically evaluate both technical and nontechnical materials.

- Adapt writing techniques that are successful for both non-technical and technical communication.

Syllabus

Listening: active and passive listening, intensive listening, developing effective listening skills, barriers to effective listening, listening to longer technical talks, listening to classroom lectures, listening to documentaries and making notes, TED talks. Exercises based on audio materials like radio and podcasts, listening to songs, practice and exercises.

Spoken: phonetics, Pronunciation practice: Problem sounds, MMFS (Multimedia Feedback System), stress, intonation, pitch, uses of a dictionary for pronunciation practice. Conversation practices.

Reading, comprehension and summarizing Reading styles, speed, valuation, critical reading, SQ3R method, PQRS method - Speed reading Comprehension: techniques, understanding textbooks, marking and underlining - Note-taking: recognizing non-verbal cues.

Formal writing - Technical writing: differences between technical and literary style - Letter writing (formal, informal and semi-formal), job applications, minute preparation, CV preparation (differences between Bio-Data, CV and Resume), and reports - Elements of style - Common errors in writing, describing a process, Statement of Purpose, instructions, checklists.

Textbooks/References

1. John Seely, The Oxford Guide to Writing and Speaking, Oxford University Press, New Delhi, 2004.
2. Turton, N.D and Heaton, J.B, Dictionary of Common Errors, Addison Wesley Longman Ltd., Indian reprint 1998.
3. Anderson, P.V, Technical Communication, Thomson Wadsworth, Sixth Edition, New Delhi, 2007.

IHSXXX Soft Skills and Professional Ethics

Course Objectives

- This course is designed to improve students' soft skills and make them aware of work ethics.
- Students should be able to take part in debates, group discussions, interviews, and presentations, making use of effective communication and presentation techniques.
- Since good communication skills help further one's career, every effort will be made to increase the students' awareness of appropriate professional techniques.

- It is also designed to develop their writing and communication skills. This course will give more focus to the practical application of each module.

Course Outcomes

- Develop soft skills
- Groom corporate habits
- Face interviews with confidence.
- Aware about the work ethics
- Create the right impression in Interviews
- Guidance for paper presentation

Syllabus

Soft Skills - SMART Goals, Teamwork, Autodidacticism, Time Management, Stress Management, Positive Attitude, SWOT Analysis.

Interview Skills - Self-Awareness, Confidence, Resume Writing, Body Language, Interview FAQs, Rejections.

Group Discussions & Mock Interviews, Practical Writing Skills- Article writing, Paper presentation for IT students, Analytical and issue based essays, Report writing: basics of report writing, structure of a report, types of reports, references, bibliography -Referencing style (IEEE Format) - Plagiarism.

Professional Ethics - Responsibilities and Rights of Professionals, Managing Multiple Roles, Public Relations.

Modern-day Research and Study Skills - Search engines and how to get optimal results using language, repositories, forums such as Git Hub, Stack Exchange, OSS communities (MOOC, SWAYAM, NPTEL), and Quora

Textbooks/References

1. Stephen E. Lucas, The Art of Public Speaking, 10th Edition; McGraw Hill Education, 2012.
2. David F. Beer and David McMurrey, Guide to Writing as an Engineer, John Wiley. New York, 2004.
3. Goodheart-Willcox, Professional Communication, First Edition, 2017.
4. Gopalaswamy Ramesh and Mahadevan Ramesh, The Ace of Soft Skills: Attitude, Communication and Etiquette for Success, Pearson Education; 1 edition, 2013.
5. Samson et.al. English for Life-4, New Delhi: Cambridge UP.